



Experimental study and kinetic analysis of the laminar burning velocity of NH_3 /syngas/air, NH_3 /CO/air and NH_3 /H₂/air premixed flames at elevated pressures

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ABSTRACT

Mixing ammonia with syngas can be a promising way to overcome the low reactivity of ammonia, allowing it to find usage in IGCC (Integrated Gasification Combined Cycle) systems and gas turbines for power generation. However, fundamental experimental data on laminar burning velocity of NH_3 /syngas/air are rather scarce, especially at elevated pressures. This information is critical for the development and validation of reaction mechanisms and advances in combustor design. In the present work, measurements of the laminar burning velocities (S_L) of NH_3 /syngas/air, NH_3 /CO/air, and NH_3 /H₂/air premixed flames were performed by the heat flux method at pressures up to 5 atm, equivalence ratios ranging from 0.7 to 1.6, ammonia mole fractions in the fuel mixture from 0.2 to 1.0 in the NH_3 /syngas/air mixtures and 0.03–1.0 in the NH_3 /CO/air mixtures. Several recently published ammonia oxidation mechanisms were tested against the present experimental data. The measurements and predictions of S_L exhibit discrepancies especially for NH_3 /H₂/air flames at elevated pressures. The pressure exponent factors, β , characterizing burning velocity at elevated pressure via empirical power-law correlation $S_L/S_{L0} = (P/P_0)^\beta$ are extracted from the measured S_L and compared with the numerical results. The thermal, diffusion, and chemical effects of blending syngas with ammonia on S_L of the mixtures are distinguished, and the dominant role of the adiabatic flame temperature on the variation of the pressure exponent β is discussed. Kinetic modeling and sensitivity analyses showed that reactions of NH_i to N_2H_i ($i = 0-4$) species affect the predicted S_L under rich conditions. At elevated pressures, these reactions also affect the NO formation via third-body collision reactions and $\text{NH}_i + \text{NO}$ reactions. Even for rich flames, the ammonia consumption is favored with the addition of syngas which also promotes NO formation by enriching the H and OH radical pools and increasing the flame temperature. The addition of hydrogen or carbon monoxide has equally promoting effect on the ammonia decomposition and NO_x formation although their flame speed differs a lot.

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1. Introduction

As a carbon-free fuel, ammonia has recently attracted the interest in the power generation sector as a means to reduce greenhouse gases. The thermal properties of ammonia are similar to those of propane in terms of boiling temperature (-33.4°C) and condensation pressure (9.9 atm) at ambient temperature, making it attractive as a hydrogen and energy carrier [1]. In addition to the

wide application of ammonia as a chemical additive for the selective non-catalytic reduction (SNCR) of the thermal DeNO_x process in large-scale power stations, direct combustion of ammonia may also be essential to replace fossil fuels in thermal power generation due to the well-established supply chain and reduced carbon footprint.

The critical issues to fire ammonia in practical combustion devices are related to its low flame speed and long ignition delay times [2,3]. To overcome these restrictions, it was suggested to blend ammonia with light fuels, e.g., hydrogen and methane [4,5] or heavy fuels, e.g., gasoline and diesel [6,7]. These types of

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Table 1

Previous experimental studies on S_L of $\text{NH}_3/\text{CH}_4/\text{air}$, $\text{NH}_3/\text{H}_2/\text{air}$, and $\text{NH}_3/\text{H}_2/\text{CO}/\text{air}$ premixed flames performed at atmospheric and elevated pressures.

Reference	Method*	Blend	X_{NH_3}	P (atm)	ϕ
Lee et al. [21]	CVCC	NH_3/H_2	0.5–1.0	1	0.6–1.67
Kumar and Meyer [9]	CF	NH_3/H_2	0–0.38	1	0.5–1.1
Li et al. [10]	CF	NH_3/H_2	0.4–0.67	1	0.6–1.4
Ichikawa et al. [16]	CVCC	NH_3/H_2	0–1.0	1–5	1.0
Lhuillier et al. [18]	CVCC	NH_3/H_2	0.4–1	1	0.8–1.4
Han et al. [13, 14]	HFM	NH_3/CH_4	0–1.0,	1	0.7–1.6
		NH_3/H_2	0.55–1.0,	1	0.7–1.6
		NH_3/CO	0.03–1.0,	1	0.7–1.6
		$\text{NH}_3/(5\% \text{H}_2 + 95\% \text{CO})$	0–1.0,	1	0.7–1.6
		$\text{NH}_3/(50\% \text{H}_2 + 50\% \text{CO})$	0.3–1.0	1	0.7–1.6
Okafor et al. [22]	CVCC	NH_3/CH_4	0–0.52	1–5	0.7–1.3

* CVCC – constant volume combustion chamber, CF – conical flame, HFM – heat flux method.

mixtures have already been successfully applied in gas turbines and engines [5,8]. However, syngas, composed mainly of hydrogen and carbon monoxide, is often used as the primary fuel in the integrated gasification combined cycle (IGCC) processes but has rarely been investigated to increase the combustion reactivity of ammonia. Furthermore, fundamental researches on laminar burning velocities (S_L) of ammonia blended with syngas are rather scarce, especially at elevated pressures. The literature studies on S_L of ammonia blended with light fuels are summarized in Table 1. The measurements of S_L were mainly based on spherically propagating flames and conical flames but have some drawbacks. Kumar and Meyer [9] reported that the heat losses from the Bunsen burner lead to the decrease of the measured S_L values, and Li et al. [10] noted that the measured S_L using conical flames are affected by flame stretch effects. While the S_L data obtained for conventional fossil fuels from spherical flames can be corrected for flame stretch effects, for ammonia flames the results are greatly influenced by the strong buoyancy effects and instabilities due to the low ammonia burning velocity [11,12]. Blending ammonia with fuels with higher S_L , increasing the oxygen content or increasing the initial temperature are possible solutions [13–18] to overcome these buoyancy effects. The recently established heat flux method under high pressures can provide adiabatic, stretch-less and buoyancy independent flames [19,20], and this is an attractive technique for the measurement of slow laminar burning velocities like those associated with ammonia flames [13,14].

The literature studies consistently show that blending NH_3 with CH_4 , H_2 or H_2/CO increases the flame speed of the fuel mixture regardless of the equivalence ratio ($\phi = (F/A)/(F/A)_{\text{stoichiometry}}$, where F is the fuel mole fraction, and A is the oxidizer mole fraction), pressure and ammonia contents. However, there are three main issues of concern from the literature studies which need further investigation. Firstly, the literature studies reveal important discrepancies between experimental data and kinetic simulations, indicating that predictions of S_L yield better agreement for $\text{NH}_3/\text{CH}_4/\text{air}$ mixtures [13,22] than for $\text{NH}_3/\text{H}_2/\text{air}$ mixtures [9,14]. This indicates that existing kinetic mechanisms for the oxidation of ammonia still have uncertainties and cannot cover different ammonia blends accurately within one mechanism, especially for ammonia/hydrogen mixtures and the reason for the disparities needs more detailed investigation. Secondly, at elevated pressures, literature studies [23,24] show that ammonia chemistry, e.g. the N_2H_x reactions are still uncertain and will affect the ammonia oxidation and ignition delay time (IDT) predictions. Moreover, some mechanisms [25,26] which can capture the IDT of the fuel-rich NH_3/H_2 mixtures cannot predict the fuel-lean mixtures as reported by He et al. [27]. This indicates that the mechanism cannot cover different equivalence ratios and different pressures accurately at the same time. Thirdly, some kinetic models like those by Duynslaegher et al.

[28], Klippenstein et al. [26], and Tian et al. [29], well predict the NO formation measured by Brackmann et al. [30], but fail to predict the flame speed of ammonia/air flames [12,22,30] and ammonia/hydrogen flames [9]. This indicates that the mechanisms cannot cover both NO_x emissions and flame speed prediction at the same time. To this end, there is an urgent need for experimental data of S_L of ammonia blending fuels at elevated pressures to scrutinize the remaining uncertainties of the ammonia chemistry.

Here, our interest lies in investigating the blending effects of syngas on the combustion characteristics of ammonia. The specific aims of the present work are: (1) to obtain highly reliable S_L data and its pressure dependence for $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$ flames at pressure up to 5 atm, equivalence ratios from 0.7 to 1.6, X_{NH_3} from 0.2 to 1.0 in the $\text{NH}_3/\text{syngas}/\text{air}$ mixtures and 0.03–1.0 in the $\text{NH}_3/\text{CO}/\text{air}$ mixtures, and assess several recently published kinetic mechanisms; (2) to quantify the thermal, diffusion and chemical effects on S_L for different NH_3 based fuel mixtures; (3) to investigate the difference in flame speed and NO formation predictions caused by different mechanisms; and (4) to distinguish the chemical effects of hydrogen and carbon monoxide addition on ammonia oxidation.

2. Experiment

2.1. Experimental setup

Experiments were conducted in a home-designed high-pressure chamber equipped with a specially designed heat flux burner, which allows for the establishment of unstretched adiabatic stable flames anchored on a perforated burner plate. The experimental setup is shown in Fig. 1. The values of S_L at elevated pressures were obtained without extrapolation, as discussed in detail in our previous publication [19].

The high-pressure chamber was designed to support pressures up to 50 atm. The burner plate used has a 20 mm diameter with perforated holes of 0.3 mm diameter and a 0.4 mm pitch. The flow rates of the gases supplied to the chamber were measured with high-pressure mass flow controllers (HMFC, Brooks Instrument Co. Ltd.). A corrosion-resistant mass flow controller was used for the NH_3 stream. The pressure was controlled by a controller with a manually operated valve and an electrical PID valve, which was installed in the exhaust gas pipeline. The exhaust gases released from the top of the chamber were directed to a heat exchanger and condenser to collect the water vapor prior to feeding it to the pressure control valves.

In the heat flux method, the heat loss from the flame to the burner can be compensated by the heat gain by the unburnt gas flowing through the burner plate. When the radial temperature gradient is equal to zero, the adiabatic laminar burning velocity

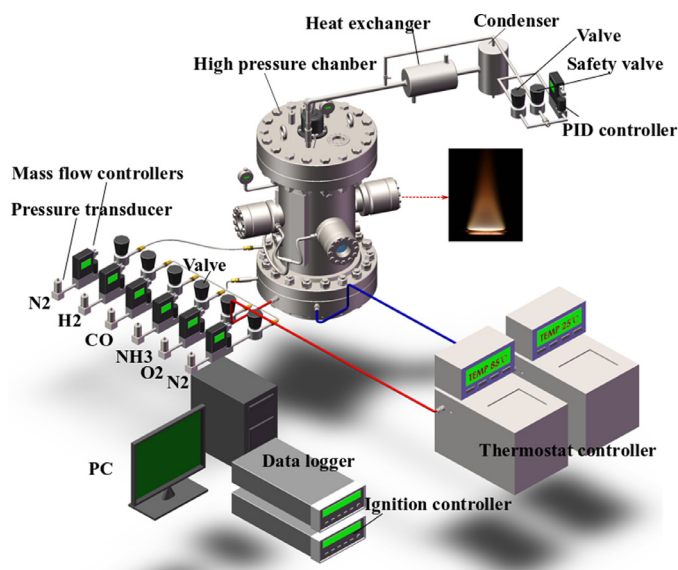


Fig. 1. Experimental setup for the high pressure laminar burning velocity measurement.

can be determined from the feeding gas velocity [31,32]. The temperature of the heat flux burner chamber was controlled constantly at 298 K using a thermostatic water bath. A thermostatic oil bath was also used to keep the burner plate temperature around 413 K to stabilize flat flames under elevated pressures. The radial distribution of the temperature over the burner plate is given by [31,32]:

$$T_p(r) = T_{center} + C \cdot r^2 \quad (1)$$

where r is the radial distance from the center of the burner plate and C is the parabolic coefficient for a given cold gas velocity v_g . When the gas feeding velocity is adjusted to obtain a constant burner plate temperature ($T_p(r) = T_{center}$), the adiabatic condition with $C = 0$ and $S_L = v_g$ is, therefore, obtained [31,32]. The heat flux method has already been applied to the S_L measurement at elevated pressures [19,20,33–37].

All gases used in the present study had purities higher than 99.99% (Jingong Gas, Hangzhou, China). In this study, ammonia was blended with three gases: syngas (50 vol.% H_2 + 50 vol.% CO), CO, and H_2 . The mole fraction of the component M in the fuel mixture is defined as $X_M = V_M/V_{fuel\ mixture}$.

2.2. Flame instability in S_L measurements

There are three formation mechanisms of flame instability: diffusional-thermal instability, which is related to the nonequidiffusion effects of thermal diffusivity (D_T) and deficient reactant mass diffusivity (D_i) when mixture Lewis number ($Le = D_T/D_i$) is less than unity, and hydrodynamic instability which is related to the density ratio and flame thickness. Cellular flame with cell size to be one order larger than the flame thickness will be experimentally observed, this happens for lean hydrogen/air and rich propane/air flames. The increase in density ratio and decrease in flame thickness will enhance the flame instability. For the limited cellular flames observed (NH_3/H_2 /air flames with $X_{NH_3} = 0.6$ for $\phi = 0.7$ and 0.8 at 5 atm), S_L was obtained from extrapolations using negative C values. As equivalence ratio increases, diffusional-thermal instability decreases while no obvious hydrodynamic instability was observed over all pressures. The present burner plate has perforated holes of 0.3 mm diameter which are small enough to efficiently exclude the local stretch at elevated pressure [33, 38] and therefore, no irregular (chaotic) distortions of the smooth

flame surface appeared. Another instability is buoyancy-driven, which occurs for fuels with low flame speed [39]. Pfahl et al. [11] and Lee et al. [21] observed obvious upward motion of the spherical flame kernel of NH_3 /air and NH_3/H_2 /air mixtures and found that the buoyancy introduces large uncertainty in flame speed measurement near the flammability limits. However, the burner-stabilized flame is downwardly propagating and the denser, unburnt mixture is below the lighter, burnt product and thus the present experiments are buoyantly stable.

The stoichiometric ammonia/air flame at 1 atm investigated here has a tilting edge at the outer of the flat flame as shown in Fig. S1, this is due to the low flame speed of ammonia and the cooling effect of the surrounding air. After the burner plate temperature T_p is increased to 413 K or further, this phenomenon is alleviated [13] without affecting the measuring results of the heat flux method [38]. Moreover, as ammonia content in the fuel decreases or pressure increases, the tilting edge is mitigated as shown in Fig. S1 because the flame front comes closer to the burner plate due to the increased mass burning rate which favors the heat exchange between flame and burner plate and makes flame edge smoother. However, small oscillation of the laminar flame structure has been observed for limited conditions like ammonia/air at 5 atm. One possible reason is that the presence of the recirculation zone of hot gases inside the high-pressure chamber induces small fluctuations of the pressure and then produces a flickering of the ammonia flame. Fortunately, the fluctuations of the measured temperature profile of the burner plate are similar for NH_3 /air flames and NH_3 /syngas/air flames with low X_{NH_3} , the latter conditions are flat flames. This means that the fluctuations of the measured temperature caused by tilting edge and the flame flickering are within the intrinsic fluctuations of the thermocouples and can be neglected.

2.3. Radiation effect on S_L measurements

Radiation heat losses and radiation absorption effects of CO_2 , H_2O , etc. have a great impact on the laminar burning velocity measurement especially at elevated pressures [40,41]. For ammonia flames Nakamura et al. [42] numerically evaluated the effects of radiation heat loss on the laminar burning velocity using an optically thin model. They found that the H_2O governed the total radiation heat loss which is important at conditions near the flammability limits without considering the radiation absorption. For syngas, previous studies [40,41,43] show that the total effect of radiation on S_L is weak in spherical flame propagation within the pressure range of 1–5 atm when considering the radiation absorption effect. Due to the decreasing effect of total radiation as S_L increases [40,41], the radiation effect may only be important for ammonia/air flames and is thus negligible for most ammonia/syngas/air flames in the present research. For the heat flux method, although radiation heat loss has an obvious effect on post-flame temperature profile [44], it has no significant role on the laminar burning velocity of methane/air [45] and, as pressure increases, the heat exchange between the flame and burner plate increases and is more favored than the heat loss to the surroundings.

2.4. The catalytic effect of the metal plate

It is well-known that ammonia is reactive with some metals like copper even at room temperature, therefore the thermocouples inside the burner plate were both glued and clamped with a small copper tube to prevent ammonia corrosion. Dyakov et al. [46] performed the experimental study of the ammonia doped ethane/air flames using a heat flux burner. They found the color of a perforated brass plate gradually changed after several days of experiments due to the degradation of the burner surface. However,

the effect of the radicals' diffusion and reaction toward the burner surface was found negligible due to the laminar burning velocity agrees well with the previous data [47] as well as the NO concentration measurement. Similar color deposits on the burner surface were found in the syngas combustion [48] and ammonia doped methane flames [49] without affecting the flame speed measurement, and Li et al. [50] suggested that the colored deposit may not be a direct product of the brass corrosion by ammonia due to the conversion ratio of ammonia to NO [49,50] agreed well with the literature data. In the pressure chamber, the only concern is that the humid atmosphere containing CO₂ will react with unburnt ammonia near the burner surface to form ammonium hydroxide (NH₃·H₂O) or ammonium bicarbonate (NH₄HCO₃). Thus, after 1 day of successive measurements, the chamber was purged to remove the residual humid flue gas and no obvious color deposits were perceptible so that this effect is negligible. The assumption of the minor influence of the burner plate on the flame structure discussed earlier [32,51] was thus experimentally confirmed here.

2.5. Experimental uncertainty

A thorough experimental uncertainty analysis was performed for the present experiments. The uncertainty of the measured S_L as discussed in [19], includes three aspects as the following: 1) the uncertainty associated with the temperature of the burner plate which are measured by multiple thermocouples; 2) the uncertainty associated with the inlet flow velocities of gases controlled by the mass flow controllers, and 3) the uncertainty associated with the pressure measurements by the pressure controller.

S_L is determined by interpolating the parabolic coefficient C that describes the temperature distribution of the burner plate at different inlet velocities, u , see Eq. (1). The measurement sensitivity s is determined as follow:

$$s = \left. \frac{dC}{du} \right|_{u=S_L} \quad (2)$$

The uncertainty of S_L associated with the temperature measurements by the thermocouples can be calculated as follow:

$$\Delta S_L^{TC} = \frac{\sigma_c}{s} \quad (3)$$

where σ_c is the average standard deviation of the thermocouple readings, which has a specific value for a given burner [52]. σ_c can be calculated as follows [53]:

$$\sigma_c = \left(\frac{\sum (T_i - C \cdot r_i^2 - T_{center})^2}{(N-2) \cdot \sum (r_i^2 - \bar{r}^2)^2} \right)^{1/2} \quad (4)$$

where N is the number of thermocouples, and T_i are the measured temperatures at radii r^i ($i = 1, 2, \dots, 8$ as eight thermocouples were used).

The uncertainties associated with the mass flow controllers can be estimated as follows:

$$\Delta S_L^{MFC} = S_L \frac{(\sum (\Delta F_i)^2)^{1/2}}{F_{tot}} \quad (5)$$

where F_{tot} is the total flow rate of all gases, and ΔF_i is the uncertainty of the i th mass flow controller. ΔF_i can be calculated as follows:

$$\Delta F_i = 0.8\% \cdot R_d + 0.2\% \cdot F_s \quad (6)$$

where R_d is the measured mass flow and F_s is the full range of the mass flow controller.

For S_L measurements at elevated pressures, there is an additional uncertainty due to the pressure controllers used in the present setup, which can be calculated as follows:

$$\Delta S_L^P = -S_L (1 + \beta') \frac{\Delta P}{P_{set}} \quad (7)$$

where ΔP is the uncertainty of the pressure controllers and β' is fitted by the commonly accepted power law:

$$S_L^m / S_L^{set} = (P^m / P^{set})^{\beta'} \quad (8)$$

where S_L^m , S_L^{set} , P^m , P^{set} are the actual and the set values of S_L and pressure, respectively. The overall uncertainty of the measured S_L can, therefore, be evaluated based on the uncertainties discussed above. The uncertainties are shown as error bars associated with the reported experimental data for the burning velocity and pressure power exponents. All experimental data obtained in the present study, together with their uncertainties, are tabulated in the Supplemental material.

3. Kinetic simulations

Simulation of the laminar burning velocities and premixed flame structure was performed with the PREMIX module in the Chemkin PRO package [54]. An adaptive re-gridding method with convergence conditions of GRAD and CURV values setting at 0.02, which ensures at least 500 grid points, was used for each test condition. Thermal diffusion (Soret effect) and mixture-averaged transport formulation were included in the computations. Four kinetic mechanisms were used: the mechanism of Han et al. [14], the mechanism of Shrestha et al. [55], the mechanism of UC-San Diego [56], and the mechanism of Stagni et al. [57]. The San Diego [22] mechanism is composed of the main hydrocarbon mechanism (2016.12.14) and the nitrogen sub-mechanism (2018.07.23). It has 68 species and 311 reactions (41 reactions for the nitrogen chemistry) and proved well in predicting S_L for NH₃/H₂/air and NH₃/CO/air flames at atmospheric pressure. The mechanism of Shrestha et al. [55] has 125 species, and 514 reactions, this mechanism is optimized for NH₃ and NH₃/H₂/CO oxidation and is validated against data for NO formation and ammonia/syngas flame speed. The Han et al. [14] mechanism adopted the H₂-CO sub-mechanism from Varga et al. [58] and includes optimized sub-mechanism for NH₃ with 35 species and 177 reactions. It was validated against recently published S_L data for NH₃/syngas/air flames at atmospheric pressure and ignition delay time data for NH₃/O₂/Ar mixtures at 30 atm [23]. The mechanism of Stagni et al. [57] has 31 species and 203 reactions. It updates the H-abstraction reaction of NH₃ and H₂NO and is validated against experimental data of ammonia oxidation in a jet-stirred reactor and a flow reactor under lean conditions (500–2000 K). Here they are referred as Han-Mech, Shrestha-Mech, UCSD-Mech and Stagni-Mech in the following discussion.

4. Results and discussion

4.1. Effect of X_{NH_3} on S_L of NH₃/syngas/air and NH₃/CO/air flames

Figures 2(a)–(c) shows the measured and predicted S_L of stoichiometric NH₃/syngas/air flames as a function of X_{NH_3} at 1, 3 and 5 atm and $\phi = 1.0$. Regardless of the pressure, the experimental and simulated results show that S_L decreases with the increasing X_{NH_3} . For a given pressure, the relative decrease percentage in S_L from $X_{NH_3} = 0$ to 1 ($(S_{L-XNH_3=0} - S_{L-XNH_3=1}) / S_{L-XNH_3=0}$) calculated by Han-Mech is almost 94.5% for 1, 3 and 5 atm; the biggest difference in the relative decrease percentage in S_L among 1, 3, and 5 atm occurs at $X_{NH_3} = 0.3$ ($(S_{L-XNH_3=0} - S_{L-XNH_3=0.3}) / S_{L-XNH_3=0}$), where they are 49%, 57%, and 61%, respectively. For a given X_{NH_3} , the relative decrease percentage in S_L from 1 to 5 atm ($(S_{L-1atm} - S_{L-5atm}) / S_{L-1atm}$) calculated by Han-Mech has maximum at $X_{NH_3} = 0.4$, it is 49%; while at $X_{NH_3} = 0.0$ and 1.0, they are 30% and 41%, respectively. It is observed that the three kinetic mechanisms can predict the experimental results regardless of X_{NH_3} . Shrestha-Mech shows relatively better prediction performance at 1 atm while Han-Mech

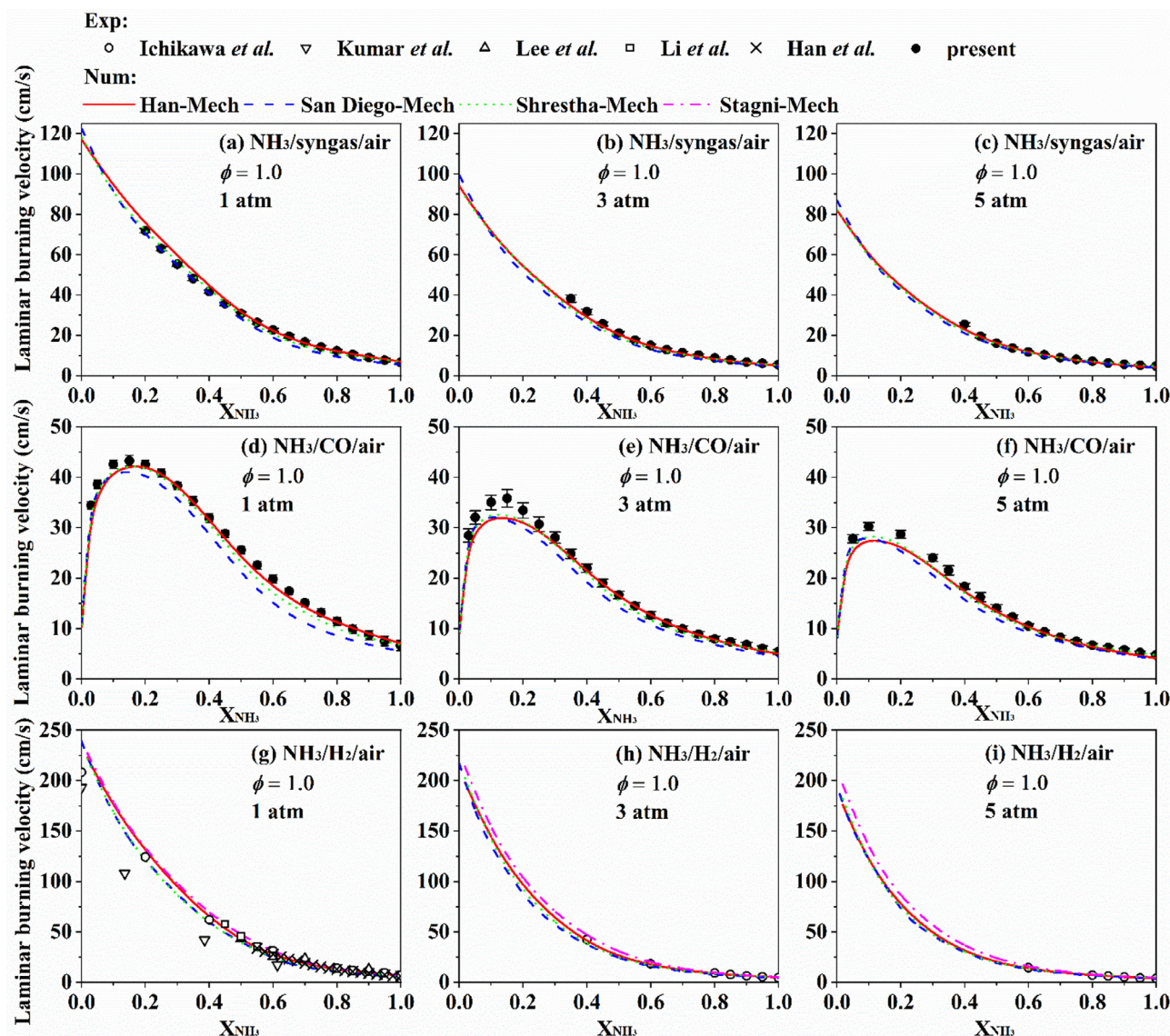


Fig. 2. Laminar burning velocities of stoichiometric $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{H}_2/\text{air}$ flames as a function of X_{NH_3} at 1, 3 and 5 atm. (Literature data are from Ichikawa et al. [16], Kumar and Meyer [9], Lee et al. [21], Li et al. [10], Han et al. [13, 14].).

slightly overpredicts the results at 1 atm but gives closer results at elevated pressures. San Diego-Mech slightly underestimates flame speeds for all pressures. Despite the scope of the mechanisms being different, the three mechanisms predict the present experimental data at both atmospheric and elevated pressures very satisfactorily.

Figures 2(d)–(f) shows the measured and predicted S_L of stoichiometric $\text{NH}_3/\text{CO}/\text{air}$ flames as a function of X_{NH_3} at 1, 3, and 5 atm. As illustrated, with X_{NH_3} increase, S_L first increases and then decreases for the three pressures. The initial rise of S_L is due to the chemical effect of the H-containing species present in the fuel mixture and here it is NH_3 . The OH radical concentration increases with the presence of NH_3 in the fuel mixture so that CO oxidation through $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$ is significantly enhanced compared to the $\text{CO} + \text{O} + \text{M} = \text{CO}_2 + \text{M}$, and the following decrease is due to the decrease of the flame temperature similar to $\text{CH}_4/\text{CO}/\text{air}$ flames reported in [59]. The experimental data reveal the occurrence of maxima in S_L approximately at $X_{\text{NH}_3} = 0.2$ for 1 atm, at $X_{\text{NH}_3} = 0.15$ for 3 atm, and at $X_{\text{NH}_3} = 0.1$ for 5 atm, indicating declining of the chemical effect of the ammonia addition on S_L as the pressure increases. Figures 2(g)–(i) shows the literature S_L data

of stoichiometric $\text{NH}_3/\text{H}_2/\text{air}$ flames at 1, 3, and 5 atm. It is found that the decrease of S_L as pressure increases or as X_{NH_3} increases is more evident in H_2 blended flames than in CO blended flames, e.g. peak S_L from Fig. 2(a)–(c) changes from 120 cm/s to 80 cm/s, while from Fig. 2(d) to (f) it changes from 45 cm/s to 30 cm/s. Predictions give better agreement in comparison with the experimental results for higher ammonia content conditions for all pressures. As ammonia content decreases, the discrepancy among measured S_L data from different literature sources at 1 atm becomes larger. The predicted S_L by different mechanisms also show more scatter for low and intermediate ammonia content $X_{\text{NH}_3} = 0.2$ –0.6. The lower values of Kumar and Meyer [9] are due to the heat loss effect of the Bunsen burner, pointed out that the corrected flame speeds agree better with the numerical results. It is found that the present measurements and predictions show good agreement for highly ammonia added conditions ($X_{\text{NH}_3} = 0.6$ –1.0) and minorly ammonia added conditions ($X_{\text{NH}_3} = 0.0$ –0.2), while the disparities are large for moderately ammonia added mixtures ($X_{\text{NH}_3} = 0.2$ –0.6). This can be explained as follows: the discrepancy between experimental and numerical S_L results is attributed to the uncertainty of the ammonia chemistry, thus the higher the net ammonia burn-

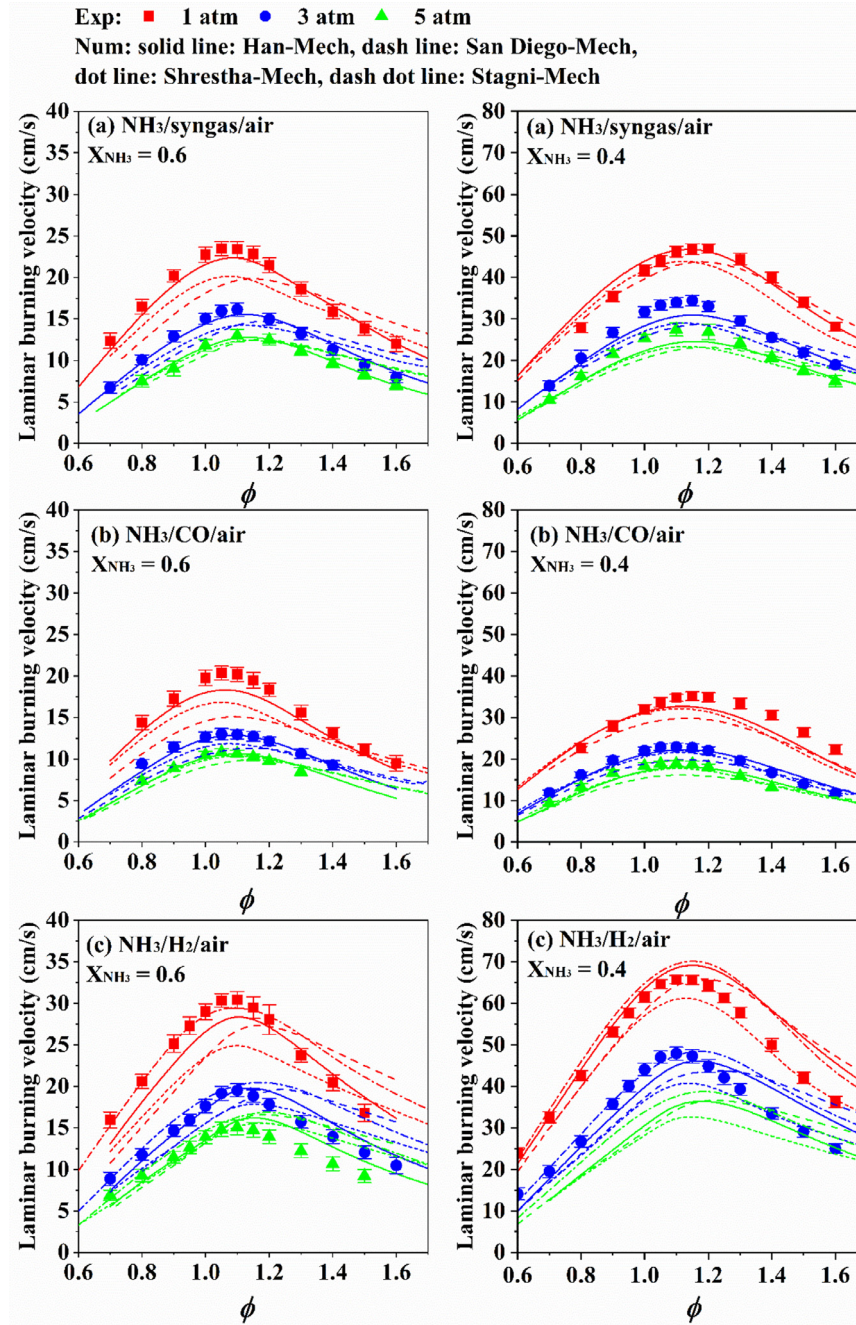


Fig. 3. Laminar burning velocities of $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$ flames as a function of equivalence ratio for various pressures and ammonia mole fractions, X_{NH_3} . (a) $\text{NH}_3/\text{syngas}/\text{air}$ at $X_{\text{NH}_3} = 0.4$; (b) $\text{NH}_3/\text{CO}/\text{air}$ at $X_{\text{NH}_3} = 0.4$; (c) $\text{NH}_3/\text{syngas}/\text{air}$ at $X_{\text{NH}_3} = 0.6$; (d) $\text{NH}_3/\text{CO}/\text{air}$ at $X_{\text{NH}_3} = 0.6$; (e) $\text{NH}_3/\text{H}_2/\text{air}$ at $X_{\text{NH}_3} = 0.6$.

ing rate in the whole mixture, the larger the discrepancies it introduces. The net ammonia burning rate depends on both X_{NH_3} and S_L ($f_{\text{NH}_3} \propto S_L \cdot X_{\text{NH}_3}$), since the previous study shows that the coupling of ammonia chemistry and hydrocarbon chemistry plays a minor role on S_L predictions [13] although it is important for intermediates formation, the oxidation of ammonia and syngas can be thought as parallel processes but sharing the common active radical pool of H and OH. At conditions approaching $X_{\text{NH}_3} = 0.0$ and 1.0, the net ammonia mass burning rate is low and contributes less to the total flame speed and uncertainties of the ammonia mechanism are not manifested, thus different mechanisms get similar predictions for pure syngas and pure ammonia as shown in Fig. 2. Thus the S_L of moderately ammonia added mixtures can better test the ammonia oxidation mechanism due to the relatively high

ammonia burning rate, for this range ($X_{\text{NH}_3} = 0.2\text{--}0.6$). Finally, the Han-Mech shows better performance for all three blends compared to other models tested.

4.2. Effect of ϕ on S_L of $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$ flames

Figure 3 shows the measured and predicted S_L of $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{H}_2/\text{air}$ flames at $X_{\text{NH}_3} = 0.6$ as a function of equivalence ratio for 1, 3, and 5 atm. It is found that the $\text{NH}_3/\text{H}_2/\text{air}$ flames have the highest peak value of S_L , followed by the $\text{NH}_3/\text{syngas}/\text{air}$, and $\text{NH}_3/\text{CO}/\text{air}$ flames. The peak S_L always occurs at $\phi = 1.1$, regardless of the pressure and the blend composition. For $X_{\text{NH}_3} = 0.6$ and all three blends, both the Shrestha-Mech

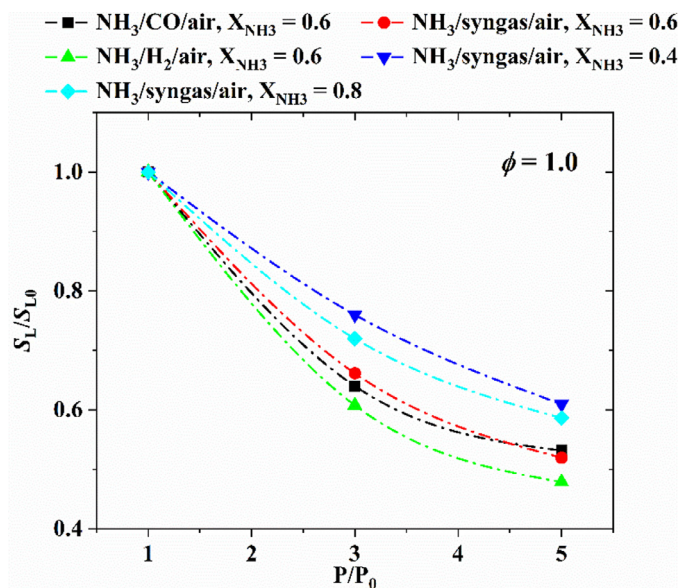


Fig. 4. Variations of S_L/S_{L0} along with P/P_0 of stoichiometric $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{H}_2/\text{air}$ mixtures.

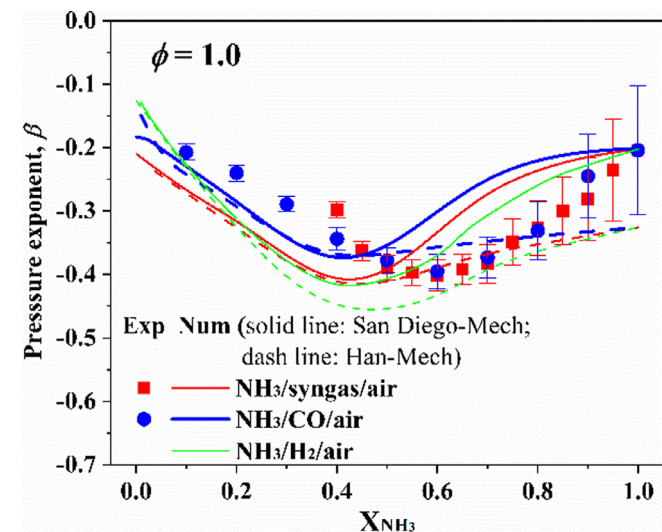


Fig. 5. Pressure exponents, β , of stoichiometric $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$ flames as a function of X_{NH_3} .

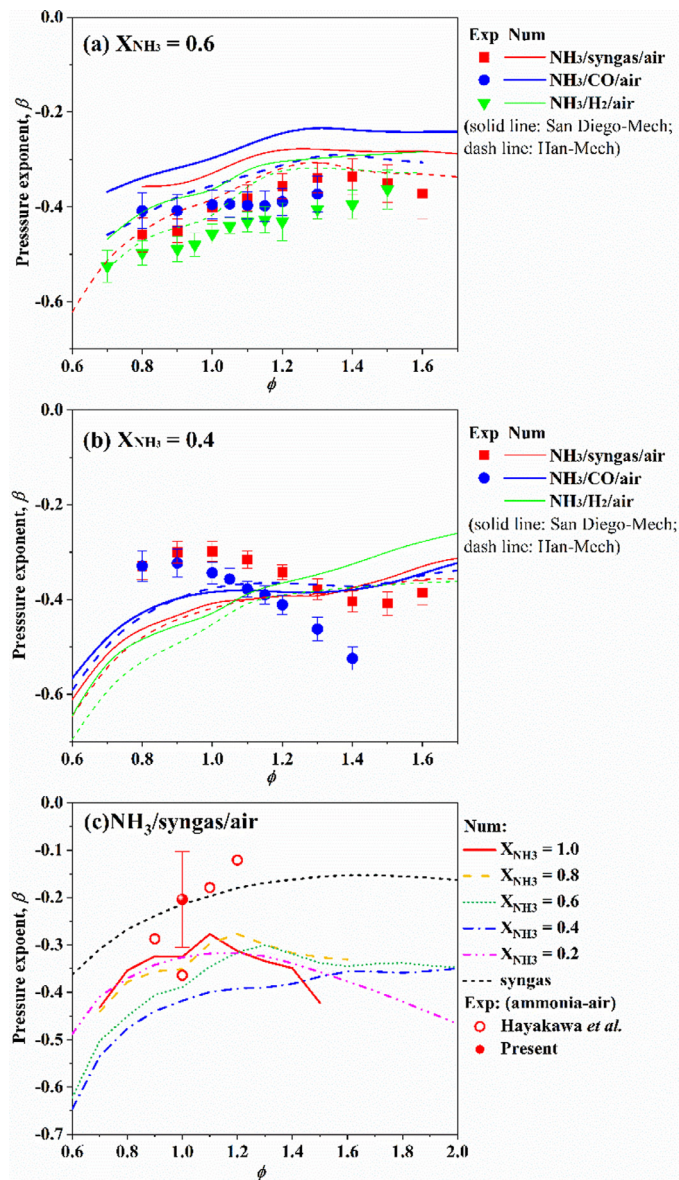


Fig. 6. Pressure exponents of $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$ flames as a function of equivalence ratio at (a) $X_{\text{NH}_3} = 0.6$; (b) $X_{\text{NH}_3} = 0.4$; (c) $\text{NH}_3/\text{syngas}/\text{air}$. (Literature data is from Hayakawa et al. [12].).

and the San Diego-Mech underpredict S_L for lean and stoichiometric flames but overpredict the S_L data for over-rich conditions at $\phi > 1.4$. In contrast, the Han-Mech gives better predictions than the other two mechanisms, especially for $\text{NH}_3/\text{syngas}/\text{air}$ flames and $\text{NH}_3/\text{CO}/\text{air}$ flames, but it still underpredicts the lean S_L of $\text{NH}_3/\text{H}_2/\text{air}$ flames at 1 atm and overpredicts the rich S_L of $\text{NH}_3/\text{H}_2/\text{air}$ flames at 3, and 5 atm. All three mechanisms' prediction accuracy increases as pressure increases except for the $\text{NH}_3/\text{H}_2/\text{air}$ flames at rich conditions, the disparities between experimental results and predictions by different mechanisms at the rich side enlarged as pressure increases.

For $X_{\text{NH}_3} = 0.4$ and all three blends, Shrestha-Mech agrees well with the experimental results at lean ($\phi < 1.0$) and over-rich ($\phi > 1.4$) side for 3, and 5 atm while underestimates flame speed at $\phi = 1.0-1.2$. San Diego-Mech better predicts the flame speed at the lean side and 1 atm while it shows larger discrepancies at the stoichiometric and rich side for 3, 5 atm. Han-Mech gives relatively better prediction at $X_{\text{NH}_3} = 0.4$ but still fails at 3 and 5

atm for $\text{NH}_3/\text{syngas}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$. All three mechanisms' prediction accuracy decreases as pressure increases at stoichiometric conditions, especially for $\text{NH}_3/\text{syngas}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$. It is concluded from the present analysis that as pressure increases, three mechanisms show deteriorating and improved agreement with the present experimental results at $X_{\text{NH}_3} = 0.4$ and 0.6 separately at lean and stoichiometric sides; while for rich conditions, all mechanism show improved prediction accuracy for the $\text{NH}_3/\text{syngas}/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames while deteriorating prediction accuracy for the $\text{NH}_3/\text{H}_2/\text{air}$ flames. Stagni-Mech better predicts the flame speed of lean $\text{NH}_3/\text{H}_2/\text{air}$ at $X_{\text{NH}_3} = 0.6$ but overpredicts it at $X_{\text{NH}_3} = 0.4$.

For both $X_{\text{NH}_3} = 0.6$ and 0.4, the discrepancies between experimental and numerical results enlarged from $\text{NH}_3/\text{CO}/\text{air}$ to $\text{NH}_3/\text{H}_2/\text{air}$, which is due to the higher ammonia burning rate at H_2 blending conditions than at CO blending conditions. The increasing net ammonia burning rate together with the uncertainty in the ammonia chemistry leads to larger uncertainty of S_L predictions and thus, the higher discrepancies in comparison with exper-

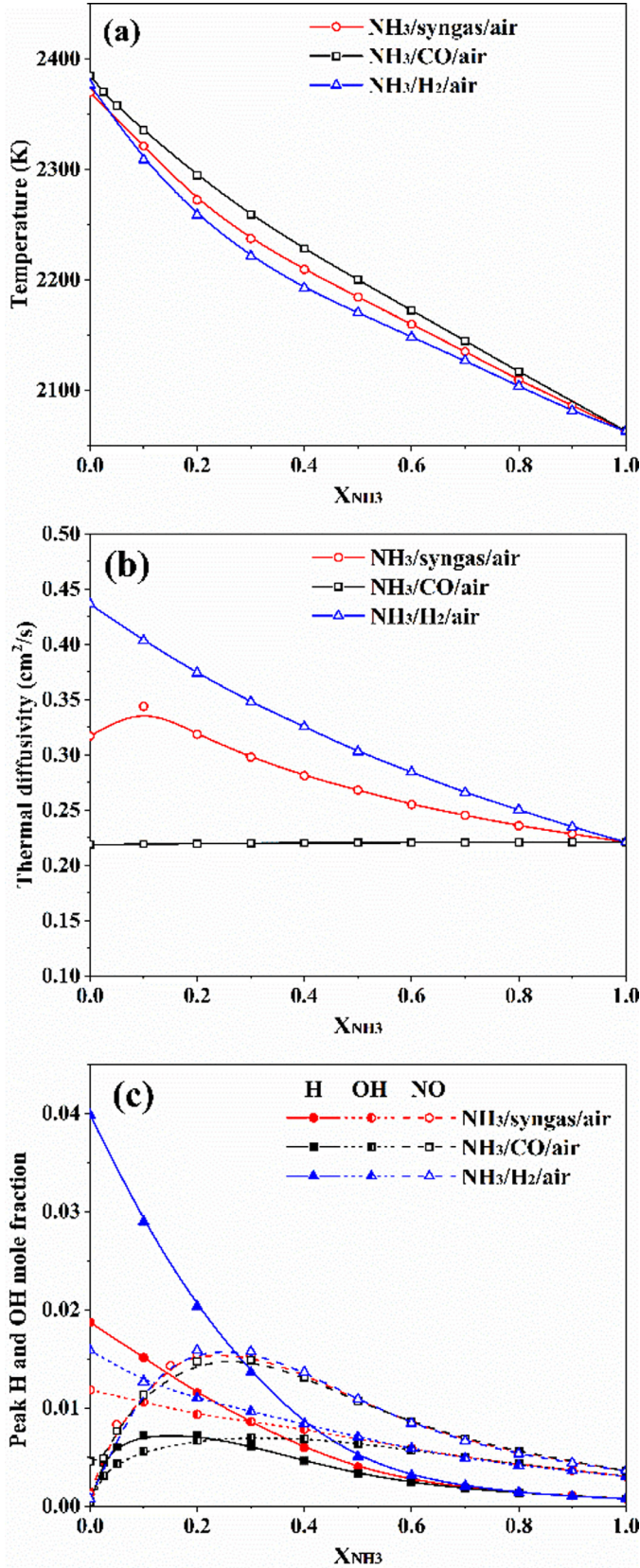


Fig. 7. Thermal characteristics of stoichiometric NH₃/syngas/air, NH₃/CO/air and NH₃/H₂/air flames as a function of X_{NH_3} at 1 atm. (a) T_{ad} ; (b) thermal diffusivity; (c) peaks of OH and H radical mole fractions calculated using the Han-Mech.

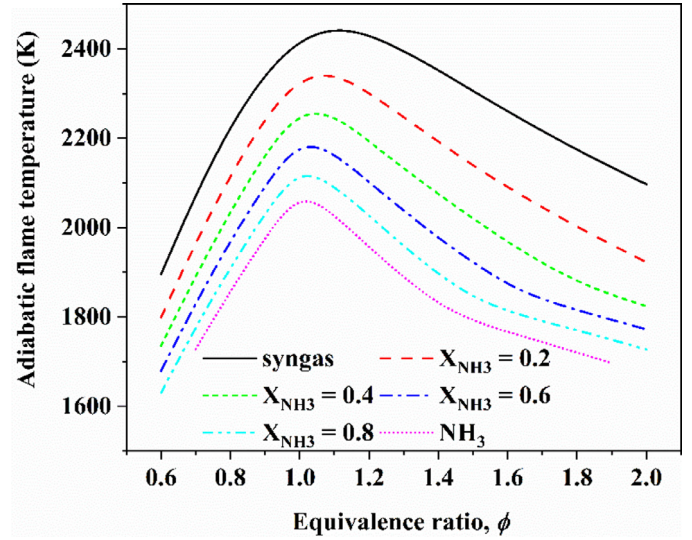


Fig. 8. Adiabatic flame temperature of NH₃/syngas/air flames at different X_{NH_3} as a function of equivalence ratio at 5 atm.

iments. Although the Han-Mech provides better predictions than the other studied mechanisms, there is still space for further improvements especially for NH₃/H₂/air flames at elevated pressures.

4.3. Effect of pressure on S_L of NH₃/syngas/air, NH₃/CO/air, and NH₃/H₂/air flames

Based on the effect of pressure on S_L , an empirical power-law correlation is obtained: $S_L/S_{L0} = (P/P_0)^\beta$ at 298 K and pressure between 1 and 5 atm, where S_{L0} is the unstretched laminar burning velocity at pressure P_0 . The pressure exponent, β was extracted from the measured and predicted laminar burning velocities. One should note that β is related to the overall reaction order $\beta = n/2 - 1$, which is around -0.39 for stoichiometric methane/air flames [19]. Figure 4 shows that S_L/S_{L0} of $X_{\text{NH}_3} = 0.6$ possesses the most significant decrease rate as P/P_0 increases among the three ammonia blends. The decrease rates of S_L/S_{L0} are similar for NH₃/syngas/air, NH₃/CO/air, and NH₃/H₂/air mixtures at fixed X_{NH_3} .

Figure 5 shows the pressure exponents, β of stoichiometric NH₃/syngas/air, NH₃/CO/air, and NH₃/H₂/air flames as a function of X_{NH_3} . It is found that β first drops to a minimum and then increases as X_{NH_3} increases for the stoichiometric NH₃/syngas/air, NH₃/CO/air, and NH₃/H₂/air flames. The values of β are similar for three blended types experimentally and numerically. The two mechanisms give similar predictions of β at the initial increase of X_{NH_3} while diverge at $X_{\text{NH}_3} > 0.6$. This suggests that the pressure dependence of S_L is primarily related to the percentage of NH₃ in the blends, X_{NH_3} rather than the syngas composition. The pressure exponents β of stoichiometric NH₃/syngas/air, NH₃/CO/air, and NH₃/H₂/air flames have experimental minima at $X_{\text{NH}_3} = 0.6$ and numerical minima at $X_{\text{NH}_3} = 0.4$. It is interesting to note that although the S_L of NH₃/CO/air flames presents a maximum at $X_{\text{NH}_3} = 0.1-0.2$, depending on the pressure, its overall reaction order changes monotonically over this range.

Figure 6(a) and (b) shows the pressure exponents of NH₃/syngas/air, NH₃/CO/air and NH₃/H₂/air flames with $X_{\text{NH}_3} = 0.6$ and 0.4 as a function of equivalence ratio. For both $X_{\text{NH}_3} = 0.6$ and 0.4, the experimental results show that the β increases slightly under lean conditions, and remains almost constant under stoichiometric conditions, but the experimental and predicted β results show divergent trend at rich conditions for three fuel types especially for $X_{\text{NH}_3} = 0.4$ conditions. Due to the relatively

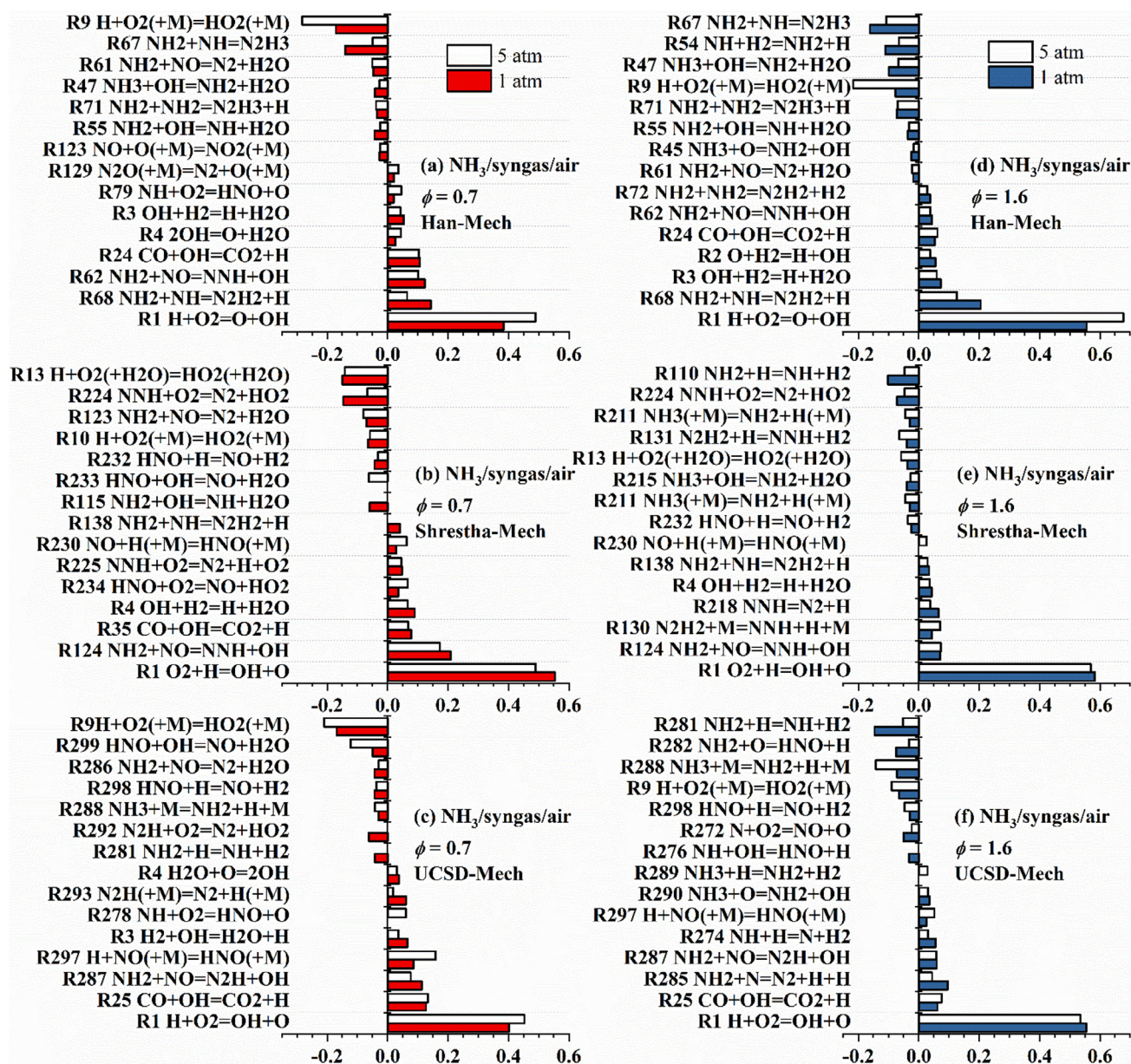


Fig. 9. Normalized sensitivity coefficients for S_L of $\text{NH}_3/\text{syngas}/\text{air}$ flames at $X_{\text{NH}_3} = 0.6$, and $\phi = 0.7$ and 1.6 , calculated using the mechanisms of Han-Mech, Shrestha-Mech and San Diego-Mech.

high ammonia content in the fuel mixture at $X_{\text{NH}_3} = 0.6$, β values are similar for the three blends, with an average value of around -0.41 . The kinetic mechanisms overpredict the pressure exponents for all fuel mixtures, regardless of ϕ . For $X_{\text{NH}_3} = 0.4$, the $\text{NH}_3/\text{syngas}/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames possess close β values under stoichiometric and lean conditions, which is similar to the observations for the corresponding flames with $X_{\text{NH}_3} = 0.6$. As ϕ increases further, the pressure exponents of the $\text{NH}_3/\text{CO}/\text{air}$ flames decrease and show larger disparities in comparison with $\text{NH}_3/\text{syngas}/\text{air}$ flames. The obvious decrease of β values at rich conditions of $\text{NH}_3/\text{CO}/\text{air}$ flames at $X_{\text{NH}_3} = 0.4$ may be attributed to the stronger collisional efficiency of CO_2 in third-body reaction $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$. It is, however, expected that with further decreasing X_{NH_3} , β values of $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{H}_2/\text{air}$ flames will get close again since the initial values of β coincide for these blends at $X_{\text{NH}_3} = 0$ (cf. Fig. 5).

Figure 6(c) shows the predicted pressure exponents of $\text{NH}_3/\text{syngas}/\text{air}$ flames using Han-Mech as a function of equivalence ratio for various ammonia contents. Under stoichiometric and lean conditions, the minimum values of β occur at $X_{\text{NH}_3} = 0.4$ as also found in Fig. 5, while under rich conditions, the minimum seems to occur at $X_{\text{NH}_3} = 0.2$ which is also confirmed by Shrestha-Mech in Fig. S2. This means that the intermediate ammonia content ($X_{\text{NH}_3} = 0.2-0.6$) is more sensitive to the pressure variation than the high and low ammonia content conditions and introduces larger uncertainty in predicting these flame speeds at elevated pressures. This explains the reason why most deviations between the measured and simulated S_L results occur at intermediate X_{NH_3} as shown in Figs. 2–3. At these conditions, the ammonia and syngas chemistry together lead to the decrease of the overall reaction rate and syngas chemistry plays a major role due to the strong third-body collision termination role of H_2O and CO_2 . Also, because

the flame speed of syngas at $\phi = 0.6$ – 2.0 is still increasing, its pressure exponent is thus still increasing and higher than for ammonia. However, Han-Mech predicts lower β of ammonia/air than the present experiments and the prediction is closer to Hayakawa et al. [12].’s results while Shrestha-Mech (cf. Fig. S2) and UCSD-Mech (cf. Fig. 5) give closer predictions in comparison with the present experiments. Han-Mech also predicts somehow the lowest β values of ammonia/air among all blends at the rich side (cf. Fig. 6) while Shrestha-Mech gives more reasonable predictions although the prediction performance for S_L of the former is better than the latter (cf. Figs. 2 and 3). This means that the pressure exponent β may be served as an independent target for mechanism validation and improvement because they give different prediction accuracy for S_L and β . The previous study [35] showed that the sensitivity spectra of the S_L and of the β coefficients are quite different, i.e. some reactions which are sensitive for S_L do not affect the pressure dependence which also confirms this point.

4.4. Thermal, diffusion and chemical effects

Figure 7 shows the adiabatic flame temperature (T_{ad}), the thermal diffusivity, and the peak mole fractions of the OH and H radicals, calculated using the Han-Mech, for NH_3 /syngas/air, NH_3 /CO/air and NH_3 /H₂/air flames at $\phi = 1.0$ as a function of X_{NH_3} at 1 atm. The results show that, as X_{NH_3} increases, T_{ad} decreases for the three blends of NH_3 , while the thermal diffusivity shows distinct trends. The NH_3 /CO/air flames always have the highest values of T_{ad} , regardless of X_{NH_3} , followed by the NH_3 /syngas/air and the NH_3 /H₂/air flames. However, the changes in the values of S_L of these three fuel mixtures do not correlate with the corresponding changes in the values of T_{ad} , which means that the thermal effect cannot explain the differences observed. Besides, it is found that, as X_{NH_3} increases, the thermal diffusivity of the NH_3 /H₂/air flames decreases, while that of the NH_3 /CO/air flames almost stays invariant at $0.22 \text{ cm}^2/\text{s}$, and that of the NH_3 /syngas/air flames increases up to $0.34 \text{ cm}^2/\text{s}$ at $X_{NH_3} = 0.1$ and then decreases. The thermal diffusivity of the flames follows the decreasing order NH_3 /H₂/air > NH_3 /syngas/air > NH_3 /CO/air similar to the trend of S_L . However, the thermal diffusivity of the flames cannot explain the variation of S_L with X_{NH_3} , especially in the case of the NH_3 /CO/air flames.

The peak mole fractions of the H and OH radicals (cf. Fig. 7(c)), which are the most important chain-branching reaction carriers, show decreasing trend along X_{NH_3} in the NH_3 /H₂/air and NH_3 /syngas/air flames but show increasing trend in the NH_3 /CO/air flames till $X_{NH_3} = 0.2$ – 0.4 and then decrease. The curves of both radicals are similar to the S_L variations with X_{NH_3} . In addition, the location of the H radical peak correlates reasonably with the location of the peak S_L of the NH_3 /CO/air flames illustrated in Fig. 2. Thus, the non-monotonic variation of S_L of the NH_3 /CO/air flames shown in Fig. 2 is due to the enrichment of H and OH radicals through the initial increase of ammonia content from zero, and the subsequent decrease is due to the reduction of T_{ad} . The shift of the peak S_L to lower X_{NH_3} side of the NH_3 /CO/air flames shown in Fig. 2 as pressure increases is due to the depletion of the active radicals H and OH which reduces the chemical enhancing effect. It is also observed that for $X_{NH_3} = 0.5$ – 1.0 , the radical mole fractions of the three fuel mixtures are similar, which leads to similar S_L values. In general, the chemical effect dominates the variation of S_L for the NH_3 /syngas/air flames, and the diffusion effect enlarges the differences.

Burke et al. [60] demonstrated that the adiabatic flame temperature defines the pressure dependence of the mass burning rate of syngas. Due to the almost identical T_{ad} of NH_3 /syngas/air flames, NH_3 /CO and NH_3 /H₂ flames, as shown in Fig. 7(a), the pressure exponents of the different fuel mixtures shown in Fig. 6(a) and (b) are almost identical; likewise, the large variation

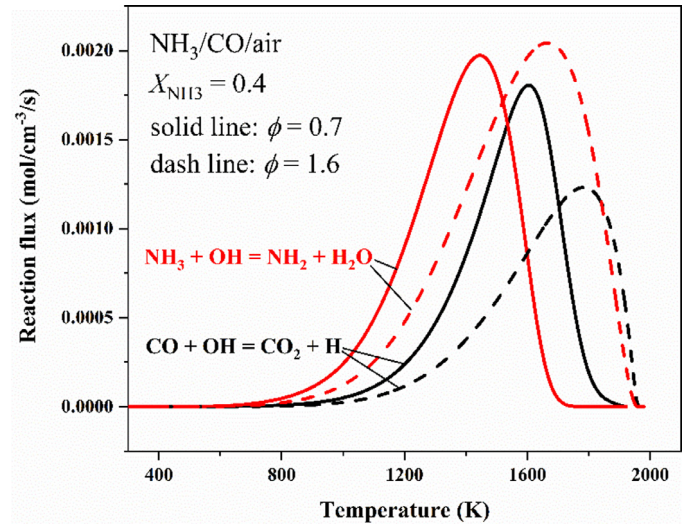


Fig. 10. Reaction flux of ammonia and CO decomposing reactions of NH_3 /CO/air flames at $X_{NH_3} = 0.4$, $\phi = 0.7$ and 1.6 , $P = 1 \text{ atm}$.

of T_{ad} of NH_3 /syngas/air flames with different NH_3 content leads to the relatively large variation of the pressure exponents shown in Figs. 5 and 6(c). Figure 8 shows that T_{ad} of NH_3 /syngas/air flames with different ammonia content has larger differences under rich conditions than under lean conditions, which may contribute to the larger scatter of the pressure exponents under rich conditions shown in Fig. 6(a)–(c).

4.5. Kinetic and sensitivity analysis

4.5.1. NH_3 /syngas/air flames

A sensitivity analysis for S_L of NH_3 /syngas/air flames with $X_{NH_3} = 0.6$, and $\phi = 0.7$ and 1.6 were conducted using Han-Mech, Shrestha-Mech and San Diego-Mech. Figure 9(a–c) shows the top 15 reactions affecting the flame speed. As illustrated under lean conditions and 1 atm, the most sensitive reaction with negative sign is $H + O_2 (+M) = HO_2 (+M)$ by Han-Mech and UCSD-Mech, corresponding to reactions $H + O_2 + H_2O = HO_2 + H_2O$ and $H + O_2 (+M) = HO_2 (+M)$ by Shrestha-Mech. Due to the third-body nature, it will be enhanced as pressure increases for both lean and rich conditions and is the major reason for the decrease of flame speed. This leads to the lower pressure exponents β observed under lean conditions compared to rich conditions, as shown in Fig. 6, which is due to the stronger participation of reaction $H + O_2 (+M) = HO_2 (+M)$ in competition with reaction $H + O_2 = O + OH$ under lean conditions. Also included in the sensitivity spectra are reactions of syngas oxidation: $OH + H_2/CO = H_2O/CO_2 + H$ whose rate constants differ for different mechanisms. However, due to their similar predictions for flame speed of syngas shown in Fig. 2 and the fact that the H_2 /CO sub-mechanism of Han-Mech, i.e. ELTE H_2 /CO mechanism [58] was validated successfully at high pressures [20], the discrepancies in predicting the flame speed of ammonia blends are caused by the sensitive reactions related to the N-species. As expected, the sensitive reactions participating in ammonia oxidation shown in the sensitivity spectra differ significantly for different mechanisms. The common sensitive reactions include: $NH_2 + NO = NNH + OH$ and $NH_2 + NO = N_2 + H_2O$ whose branching ratio not only affects S_L predictions but also the NO formation [1,25].



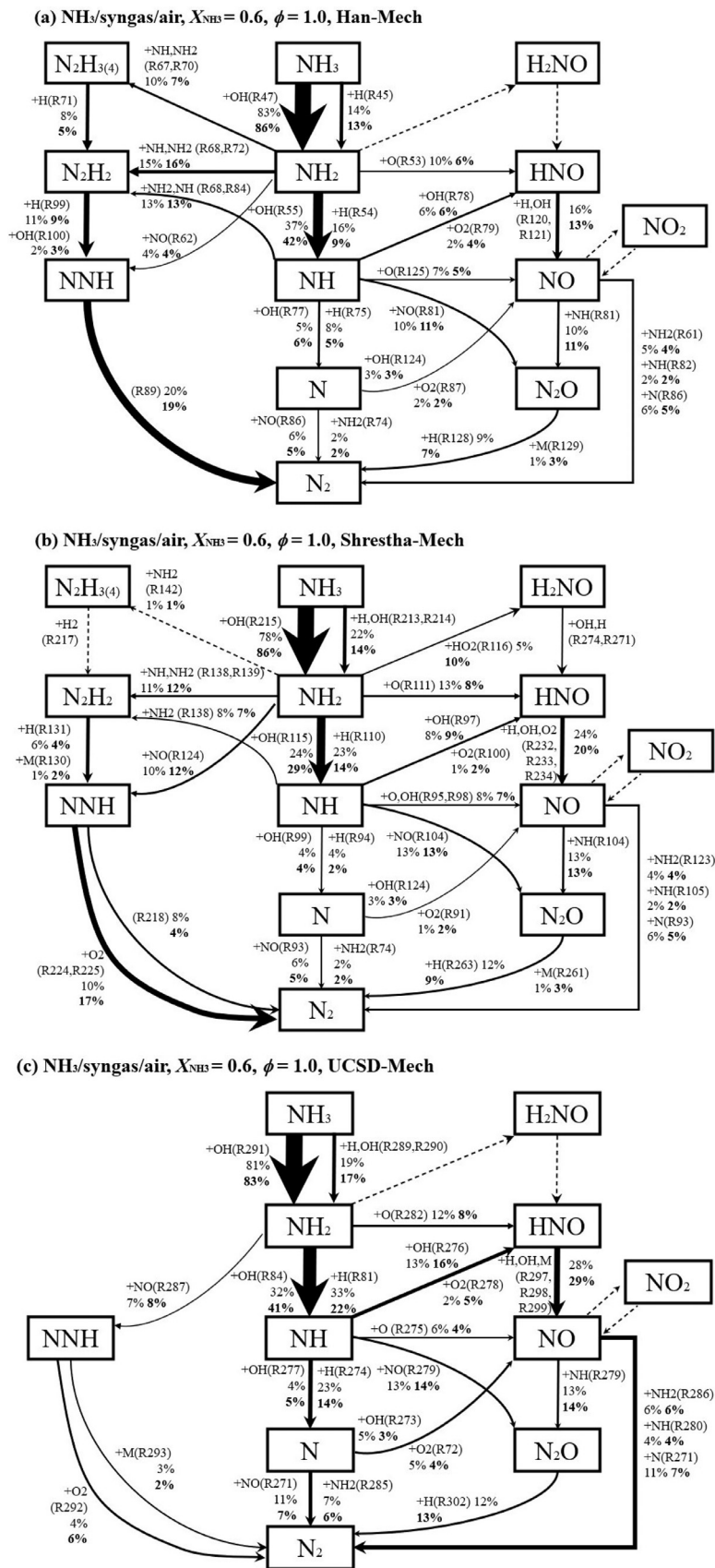


Fig. 11. Integrated rate of production analyses in the stoichiometric $\text{NH}_3/\text{syngas}/\text{air}$ flames at $X_{\text{NH}_3} = 0.6$ by (a) Han-Mech; (b) Shrestha-Mech and (c) UCSD-Mech. (The percentage and the line thickness of each pathway indicate the ratio of its reaction flux to the total reaction flux from the fuel. Regular fonts and bold fonts represent 1 and 5 atm separately.).

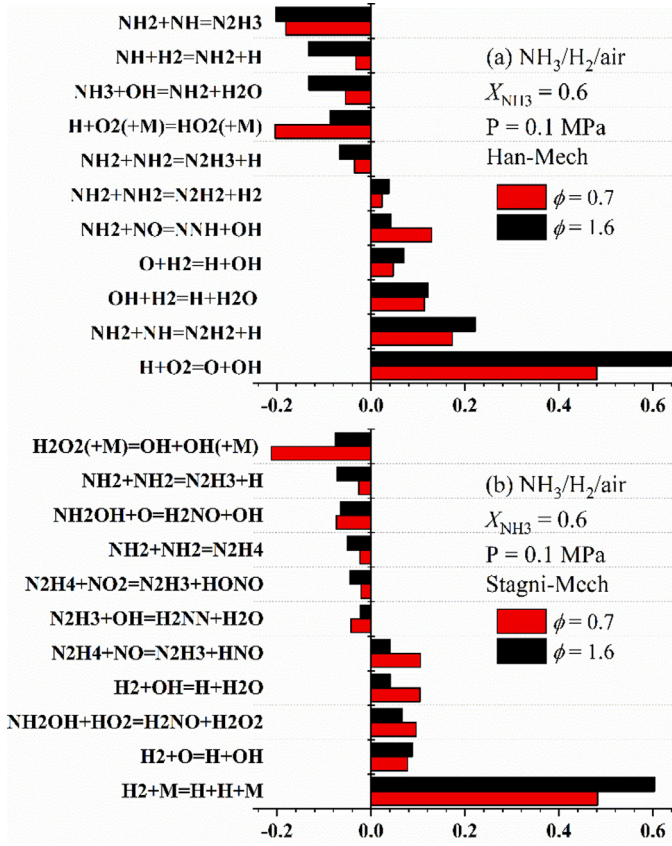


Fig. 12. Normalized sensitivity coefficients for S_L of $\text{NH}_3/\text{H}_2/\text{air}$ flames at $X_{\text{NH}_3} = 0.6$, and $\phi = 0.7$ and 1.6 , calculated using the mechanisms of Han-Mech and Stagni-Mech.

Except for these two reactions, the difference among sensitive reactions of different mechanisms is more obvious at rich conditions than at lean conditions. Under 1 atm, $\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$ is one of the most important retarding reactions in rich conditions. The rate constant at high temperatures still has large uncertainties [61,62] and the rate of this reaction is lower in Han-Mech than in Shrestha-Mech and UCSD-Mech, thus its sensitivity is not that significant than the other two. But the rate constant of $\text{NH}_2 + \text{OH} = \text{NH} + \text{H}_2\text{O}$ in Han-Mech is higher than in the other two mechanisms.



Additionally, Han-Mech and Shrestha-Mech both emphasized the importance of recombination reactions leading to N_2H_i while this pathway is not included in UCSD-Mech which is the cause of the failure to capture flame speed at rich conditions. Han-Mech and Shrestha-Mech point out the promoting reaction: $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_2 + \text{H}$ and also another retarding reaction sharing the same reactants: $\text{NH}_2 + \text{NH} = \text{N}_2\text{H}_3$ which are the most sensitive N-related reactions of Han-Mech at the rich side.



Obviously, N_2H_i reactions play a more important role in the Han-Mech because it also includes the other two recombination reactions: $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_3 + \text{H}$ and $\text{NH}_2 + \text{NH}_2 = \text{N}_2\text{H}_2 + \text{H}_2$ which are not shown in the sensitivity analysis of Shrestha-Mech.



It is found that ammonia chemistry plays a more important role at rich comparing to lean conditions, in affecting S_L especially via recombination reactions, which are more dependable on the composition of the blend, and thus the divergence of the pressure exponents β among different fuel mixtures is enlarged under rich conditions, as shown in Fig. 6. Nakamura et al. [42] also found the predicted ammonia/air flame speed has a large discrepancy in comparison with experimental results even considering the radiation effect and this discrepancy originates from the recombination reactions leading to N_2H_i . Fundamental understanding of these reactions is still insufficient [1,25] although their importance is realized at low-temperature ammonia oxidation [63] and high-pressure ammonia oxidation conditions [23,24]. Shrestha-Mech also points out the importance of N_2H_2 decomposition pathways which also affect the ammonia oxidation and have large uncertainties reported by Mei et al. [15]. The rate constant of R8 is higher in Han-Mech than in Shrestha-Mech while the former does not include R9. There is another important retarding reaction: $\text{NH}_3 + \text{M} = \text{NH}_2 + \text{H} + \text{M}$ which proceeds reversely and produces unburnt ammonia in Shrestha-Mech and UCSD-Mech but not included in Han-Mech.



The above ten reactions have large uncertainty in rate constants and introduce prediction discrepancies in comparison with the present experiments. The increasing discrepancies between experimental and numerical ammonia/hydrogen/air flame speed as pressure increases shown in Fig. 3 is also attributed to the increasing sensitivity of N_2H_i chemistry, which is however not modeled as pressure-dependent third-body reactions in most mechanisms, e.g. the Han-Mech used here. Thus, the flame speed prediction uncertainty at the rich side still needs further investigation, especially at high pressures. Moreover, the large divergence between experimental and simulated results for the pressure exponents of $\text{NH}_3/\text{CO}/\text{air}$ flames at the rich side shown in Fig. 6(b) may originate from the participation of CO_2 in the two third-body termination reactions: $\text{H} + \text{O}_2 (+ \text{M}) = \text{HO}_2 (+ \text{M})$ and $\text{H} + \text{OH} + \text{M} = \text{H}_2\text{O} + \text{M}$, and also from the increasing importance of ammonia chemistry which leads to more OH radicals participating in the oxidation of ammonia, e.g. $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$ than in the oxidation of CO, e.g., $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$, as shown in Fig. 10. It is clear that as ϕ increases, more OH radicals flow into the ammonia consumption reactions than into CO consumption reactions which introduces larger uncertainty in predicting S_L at rich conditions due to the large difference in ammonia chemistry by different mechanisms.

Only sensitivity analysis alone cannot reveal the important reaction pathways in ammonia oxidation because some reactions may have low sensitivity coefficients but share a large portion of N-species consumption. Figure 11 shows the reaction pathway analyses of stoichiometric $\text{NH}_3/\text{syngas}/\text{air}$ flames at $X_{\text{NH}_3} = 0.6$, $P = 1$, and 5 atm by three mechanisms. Percentages and line thickness represent the proportion of N-species consumption. It is found that three mechanisms predict similar reaction pathways. Ammonia is converted totally into NH_2 by H, O, and OH radicals, while the consumption fraction through OH exceeds 80%. The rate constant of $\text{NH}_3 + \text{OH} = \text{NH}_2 + \text{H}_2\text{O}$ in Han-Mech is higher than this in the other two mechanisms, thus this H-abstraction reac-

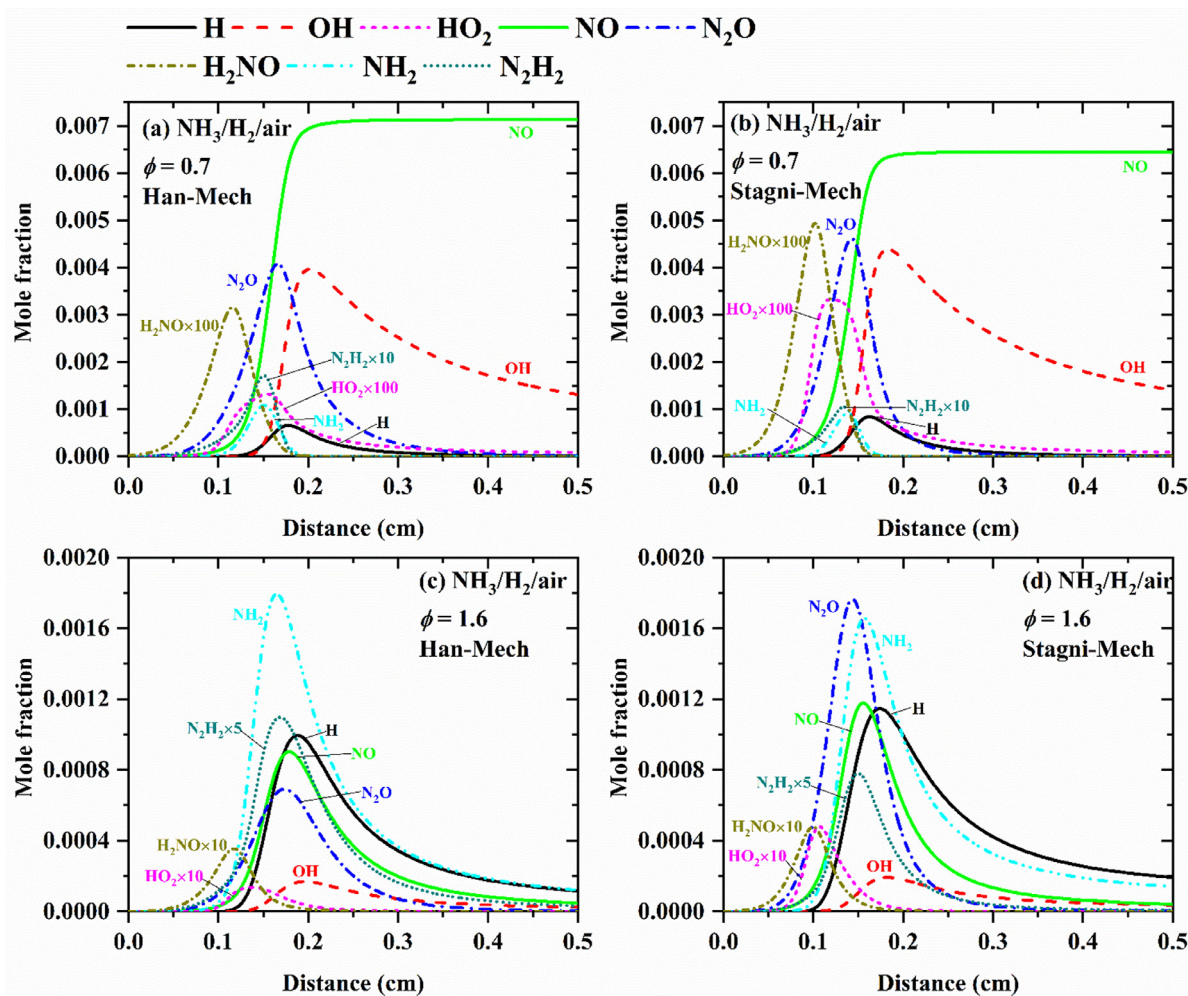


Fig. 13. Intermediates mole fraction of $\text{NH}_3/\text{H}_2/\text{air}$ flames at $X_{\text{NH}_3} = 0.6$, and $\phi = 0.7$ and 1.6, calculated using Han-Mech and Stagni-Mech.

tion always has a higher contribution in Han-Mech. The same consideration can also apply to reaction $\text{NH}_2 + \text{OH} = \text{NH} + \text{H}_2\text{O}$. Moreover, the total reaction rates of ammonia decomposition to NH_2 follow the decreasing order of Han-Mech > Shrestha-Mech > UCSD-Mech (not shown here), which coincides with the order of stoichiometric flame speed predicted by three mechanisms shown in Fig. 3. This explains why the Han-Mech gives closer S_L results to experiments.



The following difference among the three mechanisms relies in the reaction flux directions of NH_2 radicals. As the sensitivity analysis shows, Han-Mech puts more emphasis on the recombination reactions leading to N_2H_i ($i = 1-4$) and possesses the highest production of N_2H_2 . N_2H_2 is converted into NNH by H-atom abstraction reactions and finally produces N_2 . While Shrestha-Mech emphasizes the reaction pathway of H_2NO formation from $\text{NH}_2 + \text{HO}_2 = \text{H}_2\text{NO} + \text{OH}$, this step will be enhanced as pressure increases due to the increasing HO_2 mole fraction by the reaction $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2 (+\text{M})$. Most of H_2NO can lose an H atom to form HNO , while HNO totally decomposes into NO . Also due to the higher percentages of NH , NH_2 conversion to HNO and NO in Shrestha-Mech, it predicts higher NO production than Han-Mech.

Meanwhile, the difference between Han-Mech and Shrestha-Mech can also be found in the production and consumption pathway of NNH although they give similar N-species consumption fraction via NNH . Due to the complete ignorance of N_2H_i reactions, UCSD-Mech emphasizes the pathway of $\text{NH}_2\text{-NH-N-N}_2$ which contribution is also the major difference between the three mechanisms. According to the sensitivity analysis shown in Fig. 9, reaction $\text{NH}_2 + \text{H} = \text{NH} + \text{H}_2$ (R3) is the most retarding reaction in rich conditions and it owns the highest proportion in UCSD-Mech leading to the lower S_L . UCSD-Mech also weakens the NNH reaction pathway but enhances the conversion from NH , N to HNO , NO , thus the reaction pathway of $\text{HNO-NO-(N}_2\text{O)-N}_2$ has the highest contribution in UCSD-Mech.

The reaction rate through the N_2H_i route ($\text{N}_2\text{H}_i\text{-NNH-N}_2$) is always lower than the reaction rates of $\text{NH}_2\text{-NH-N-N}_2$ and $\text{HNO-NO-(N}_2\text{O)-N}_2$, e.g. the rate constants of NH_2 conversion to N_2H_i are lower than those to NH and HNO ; the rate constants of N_2 formation are always higher through N conversion, followed by NO , and N_2O conversion and the slowest is through NNH conversion. The importance of the N_2H_i route in the three mechanisms is ranked from high to low as Han-Mech > Shrestha-Mech > UCSD-Mech. Moreover, this trend will be enlarged as the equivalence ratio increases, which also explains that the trend of the flame speed reduction predicted by the three mechanisms in Fig. 3 after $\phi = 1.0$ is the highest for Han-Mech, middle for Shrestha-Mech, and the lowest for UCSD-Mech.

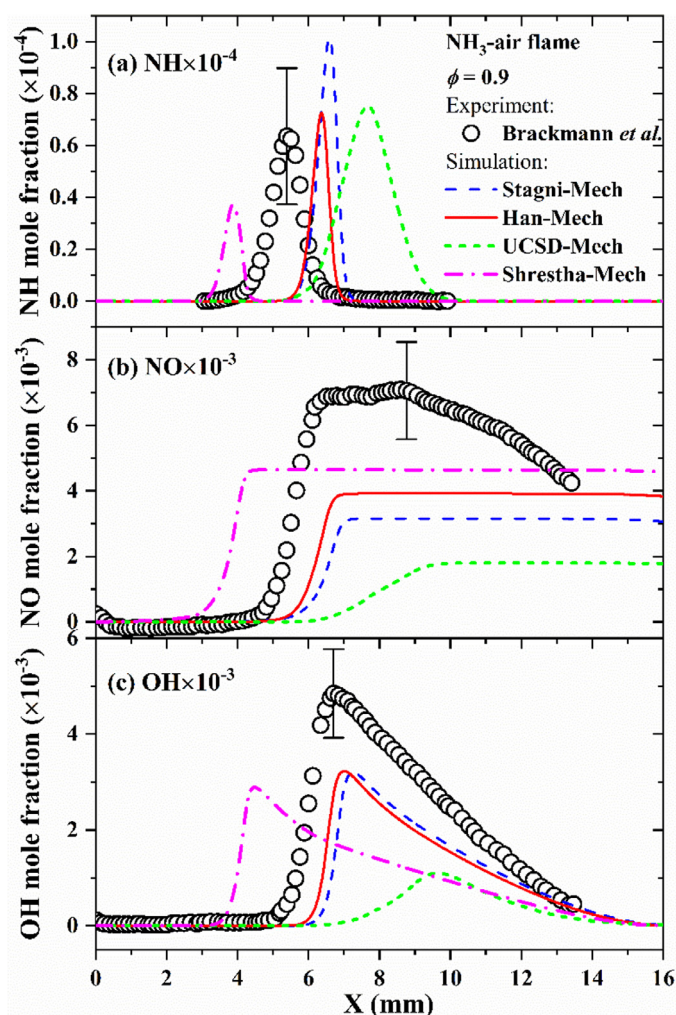


Fig. 14. Comparison of experimental measurements of species concentration on a strain-stabilized ammonia/air flames under lean condition with modeling results.

4.5.2. $\text{NH}_3/\text{H}_2/\text{air}$ flames

In Fig. 3, the most obvious disparities between experimental and numerical modeling results occur for $\text{NH}_3/\text{H}_2/\text{air}$ mixtures. Stagni-Mech showed better prediction accuracy at the lean side while Han-Mech showed better prediction accuracy at the rich side although there is still a need for improvement. To investigate which reactions define the difference between these two mechanisms, a sensitivity analysis for S_L of $\text{NH}_3/\text{H}_2/\text{air}$ flames with $X_{\text{NH}_3} = 0.6$, and $\phi = 0.7$ and 1.6 were conducted using Han-Mech, and Stagni-Mech. Figure 11(a and b) shows the top 11 reactions affecting the flame speed. Stagni-Mech predicts that the most sensitive reactions with positive and negative sign are: $\text{H}_2 + \text{M} = \text{H} + \text{H} + \text{M}$ and $\text{H}_2\text{O}_2 (+\text{M}) = \text{OH} + \text{OH} (+\text{M})$ which are different from the Han-Mech. Comparing two mechanisms, Han-Mech emphasized more the N_2H_4 recombination reactions where the NH , NH_2 , and NO radicals play an important role in the sensitivity spectra while it is not that significant in Stagni-Mech. In contrast, Stagni-Mech emphasized more the hydrogen chemistry where the H , O and HO_2 radicals play important role in the sensitivity spectra, e.g. $\text{H}_2 + \text{O} = \text{H} + \text{OH}$ and $\text{H}_2 + \text{OH} = \text{H} + \text{H}_2\text{O}$. These two reactions lead to the higher reactivity of Stagni-Mech.

The concentration of intermediates predicted by Han-Mech and Stagni-Mech of $\text{NH}_3/\text{H}_2/\text{air}$ flames at $X_{\text{NH}_3} = 0.6$ under lean and rich conditions are presented in Fig. 13. In the preheating zone, both mechanisms predict the highest concentration of

H_2NO among all species. They always increase to peak ahead of other species and then decompose into small molecules NO , H , and OH in the reaction zone. The sensitivity analysis in Fig. 12(b) by Stagni-Mech also points out the importance of H_2NO related reactions: $\text{NH}_2\text{OH} + \text{HO}_2 = \text{H}_2\text{NO} + \text{H}_2\text{O}_2$ and $\text{NH}_2\text{OH} + \text{O} = \text{H}_2\text{NO} + \text{OH}$. This is due to the fast generation of H_2NO through $\text{NH}_2 + \text{HO}_2 = \text{H}_2\text{NO} + \text{OH}$ at low and medium temperatures [57]. Thus, HO_2 and H_2NO play important role in ammonia consumption in the preheating zone and the peak concentrations of HO_2 and H_2NO predicted by Stagni-Mech is higher than that of Han-Mech leading to higher reactivity in the preheating zone of the former. As discussed above, H_2NO will mostly be converted to HNO and NO and the reaction rates of the HNO pathway in ammonia oxidation are faster than those of recombination reactions leading to N_2H_4 .

The peak concentration of N_2O predicted by Stagni-Mech is always higher than that of Han-Mech indicating more NH_2 is converted to N_2O and N_2 and the oxidation of fuel proceeds faster. However, the lower peak concentration of NO predicted by Stagni-Mech at the lean side is because N_2O shares a large portion of N-species consumption; the lower peak concentration of NO predicted by Han-Mech at the rich side is due to the higher level of NH_2 and N_2H_2 predicted by Han-Mech indicating that the reaction does not proceed as fast as the Stagni-Mech, so the NO concentration is lower and flame speed is lower.

4.5.3. Effect of syngas blending on ammonia oxidation and NO formation

Before analyzing the effect of syngas blending on NO formation, it is necessary to verify the accuracy of these four mechanisms to predict the NO formation in ammonia flames by comparing with experimental results. Brackmann et al. [30] have carried out the NO concentration measurements by laser-induced fluorescence (LIF) in a stagnation ammonia/air flames under strain-stabilized conditions and atmospheric pressure. The results were modeled and shown in Fig. 14. It is found that Shrestha-Mech gives better agreement with the experimental NO concentrations while Han-Mech gives better predictions of the peak concentration of NH and OH . The simulations used the experimental temperature profile and the convergence conditions of GRAD and CURV values of the stagnation flame model in CHEMKIN pro was set at 0.1. The decline of the experimental NO concentrations at the end of the flame may be due to the laser signal absorption of the LIF image. The difference between the peak mole fraction position of the experiments and the numerical results is due to the different mechanisms predicting the different positions of the flame front. Besides, it may be due to the difference in the temperature field and concentration field obtained by the CARS and the LIF. Consistent with the above analysis in Fig. 13, the NO concentration predicted by the Han-Mech is higher than that of the Stagni-Mech and the predicted results of the OH concentration are similar.

To investigate the effects of H_2 and CO addition on ammonia decomposition and NO generation, the species profiles in NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and syngas/air flames were calculated as shown in Fig. 15. All mechanisms give very similar fuel decomposition rates for each condition especially in the $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames, but for NO formation, predictions vary greatly. Comparing the oxidation of NH_3/air and the oxidation of syngas/air, the difference of four mechanisms is greater for the former indicating the ammonia chemistry has a larger difference than syngas chemistry. UCSD-Mech predicts the slowest ammonia decomposition rate and thus the lowest peak concentration of NO and H_2 , which is consistent with the finding in Fig. 14. The ammonia decomposition rate predicted by Shrestha-Mech is similar to the Han-Mech and Stagni-Mech, but its NO concentration is always the highest. With the addition of hydrogen or carbon monox-

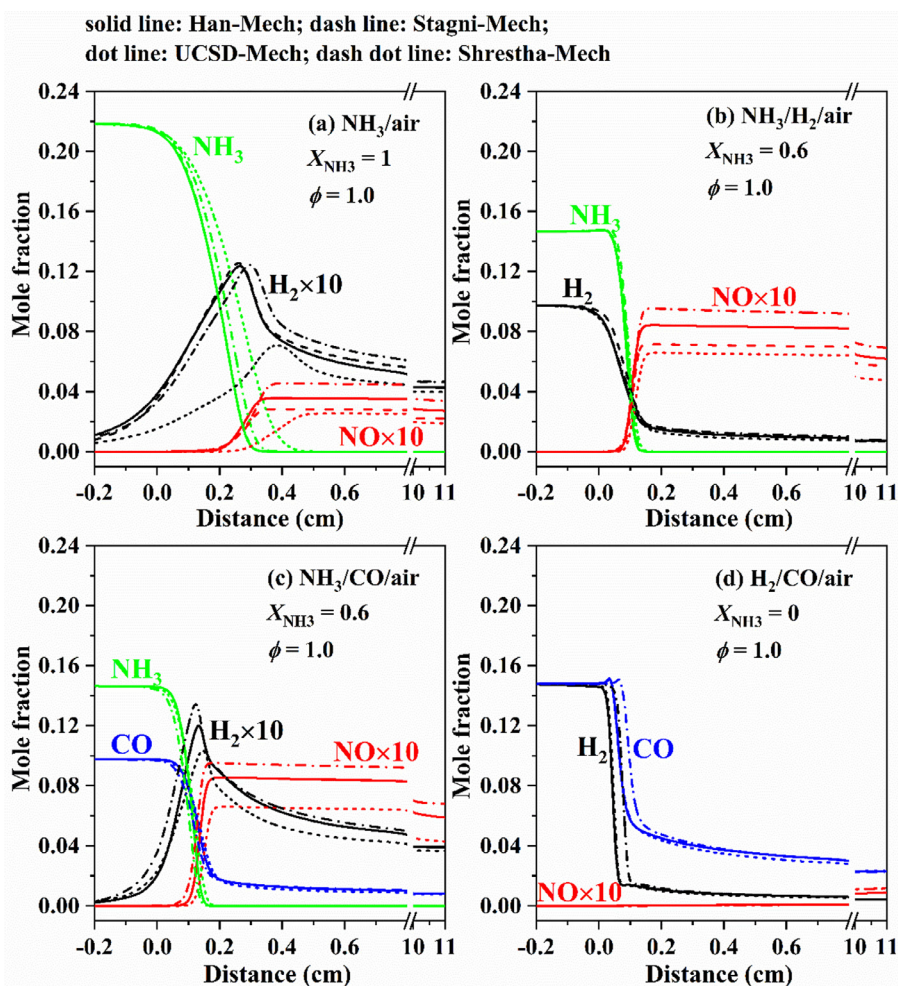


Fig. 15. Species mole fraction of stoichiometric (a) NH_3/air ; (b) $\text{NH}_3/\text{H}_2/\text{air}$; (c) $\text{NH}_3/\text{CO}/\text{air}$ and (d) $\text{H}_2/\text{CO}/\text{air}$ flames calculated using the Han-Mech, Stagni-Mech, Shrestha-Mech and UCSD-Mech at 1 atm.

ide, the ammonia decomposition rate is significantly increased and the generation of NO is greatly strengthened.

It is worth noting that although the flame speeds of $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ differ a lot (~ 10 cm/s), their peak and tail concentrations of NO are similar, and this trend is confirmed by different mechanisms. Moreover, in Section 4.4, Figure 7(c) shows that for different X_{NH_3} contents, the peak NO concentrations of H_2 or CO blending conditions are still similar. This may be due to the similar adiabatic flame temperature leading to a similar ammonia decomposition process. Figure 16 shows the emission characteristics of $\text{NH}_3/\text{syngas}/\text{air}$ mixtures at the flame tail of the calculation domain as a function of the equivalence ratio. It is found that the final NO_x (NO, N_2O , and NO_2) concentrations for H_2 or CO blending conditions are still similar under different equivalence ratios and always have a maximum value at $\phi = 0.9$. The highly reactive hydrogen blending makes the oxidation of ammonia faster and thus the concentration of NH_2 is lower than that of CO blending, but the enrichment of H radical promotes the formation of N_2H_2 .

Figure 17 presents the sensitivity analysis of NO formation of the four conditions in Fig. 15 by Shrestha-Mech and Han-Mech. The sensitivity analysis is performed at the locations of the maximum NO concentration in Fig. 15, which corresponds to about 2000 K in the reaction zone for NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$ flames, and to the end of the reaction zone for $\text{H}_2/\text{CO}/\text{air}$ flame. The thermal NO formation in syngas/air depends on the reaction $\text{N} + \text{NO} = \text{N}_2 + \text{O}$. The most sensitive reaction to NO

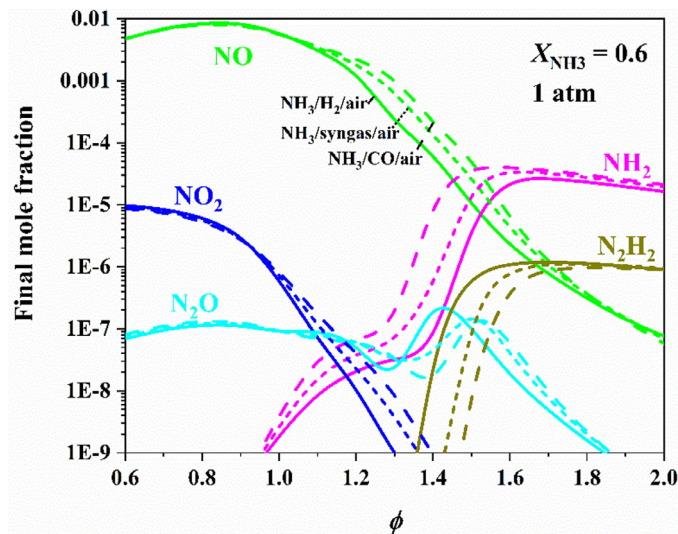


Fig. 16. Final species mole fraction of $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{syngas}/\text{air}$ flames calculated using the mechanisms of Han-Mech.

generation is the same as to the flame speed: $\text{H} + \text{O}_2 = \text{O} + \text{OH}$. The reactions that promote the NO formation include the HNO formation: $\text{NH}_2 + \text{O} = \text{HNO} + \text{H}$; the reactions that inhibit the NO gener-

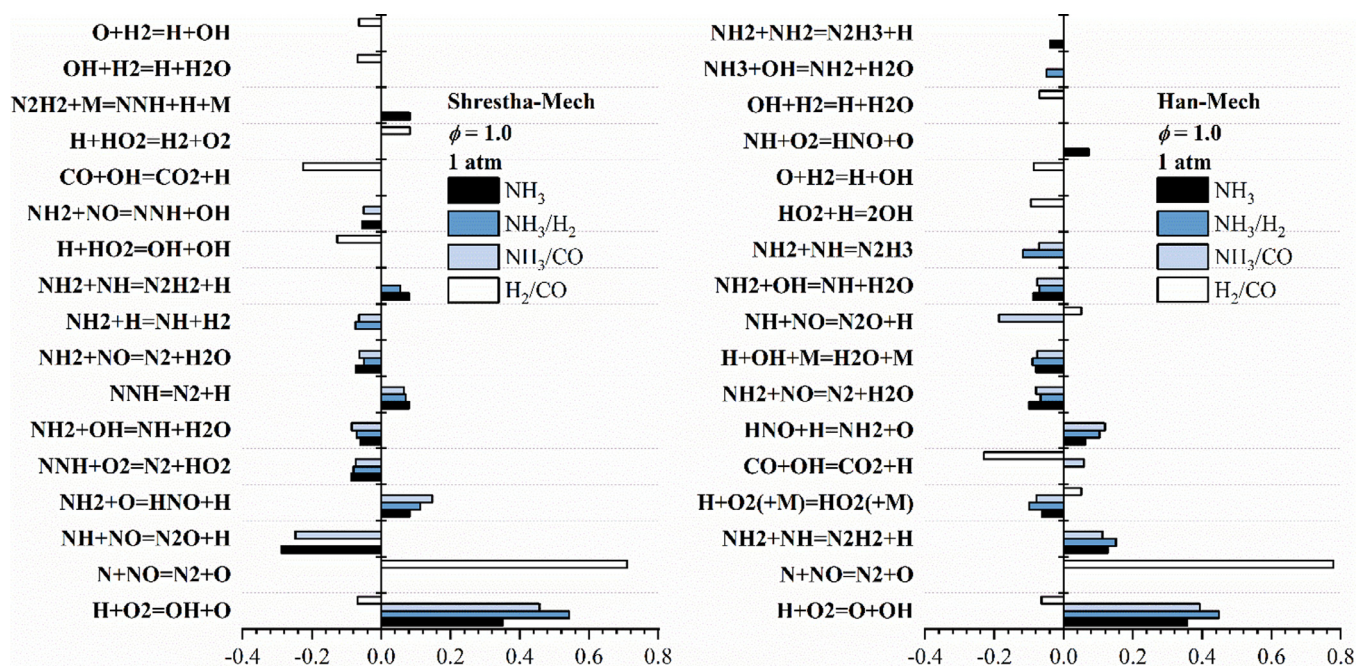


Fig. 17. Normalized sensitivity coefficients for NO generation of NH_3/air , $\text{NH}_3/\text{H}_2/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{H}_2/\text{CO}/\text{air}$ flames calculated using Han-Mech and Shrestha-Mech.

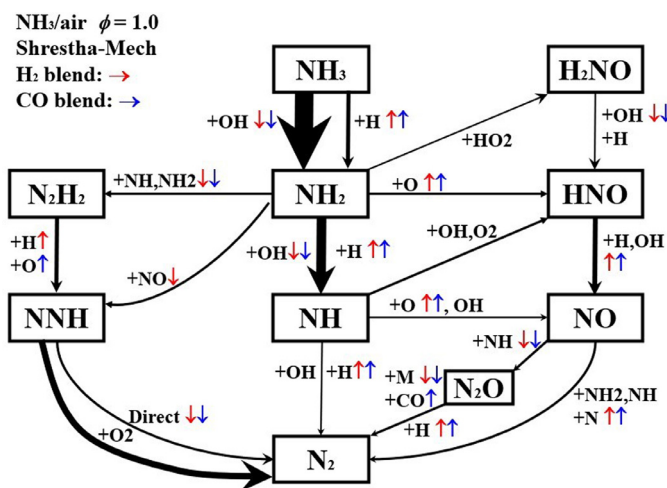


Fig. 18. Reaction pathway analysis in the stoichiometric NH_3/air flames with addition of H_2 and CO by Shrestha-Mech at 1 atm.

ation include the direct consumption of NO by NH , NH_2 as well as NH_2 conversion to NH which consumes the active radicals H and OH . Both mechanisms reveal the importance of the above reactions but their sensitivity coefficients differ a lot. Combining the aforementioned reaction pathway analyses in Fig. 11, it is found that in the Shrestha-Mech, more NH_2 reacts with O or through H_2NO route to produce HNO and finally to form NO ; while in the Han-Mech, more NH_2 is converted to NH and N_2H_i , and the N_2H_i reactions also take most seats in the sensitivity spectra shown in Fig. 17. Furthermore, the calculated NO consumption pathways and contributions for the two mechanisms are similar. These two factors lead to higher NO production of Shrestha-Mech than of Han-Mech, however, its excessive NO consumption may be the cause of the deviation from the experimental results shown in Fig. 14 and needs further investigation.

To distinguish the respective chemical effects of H_2 and CO addition on ammonia oxidation and NO formation, reaction pathway

analysis of stoichiometric ammonia/air flame in comparison with H_2 and CO addition is presented in Fig. 18. It is found that due to the addition of H_2 and CO , the fraction of ammonia in the fuel is reduced, and the chemical role of NH , NH_2 radicals is weakened, i.e. NO consumption through reaction with NH , NH_2 and NH_i recombination reactions are weakened. Sensitivity analysis of Fig. 17 shows that H_2 and CO enrich the H radical pool through $\text{OH} + \text{H}_2/\text{CO} = \text{H}_2\text{O}/\text{CO}_2 + \text{H}$ and further increases the concentration of O atom through $\text{H} + \text{O}_2 = \text{O} + \text{OH}$. Thus, they mainly increase the importance of H radicals in ammonia conversion into NH_2 , NH , and the importance of O radical in NH_2 , NH conversion to HNO , and NO . As Fig. S3 shows, the HNO decomposition pathway is enhanced with syngas addition and although the reaction rate and flame speed are higher in H_2 blending condition than in CO blending condition, they have the same rate of production of NO due to their same role in promoting the oxidation pathway of ammonia.

5. Conclusions

This work presents the laminar burning velocities of $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$ and $\text{NH}_3/\text{H}_2/\text{air}$ mixtures at $\phi = 0.6$ – 1.6 , pressures 1–5 atm, ammonia mole fraction $X_{\text{NH}_3} = 0.2$ – 1.0 in the $\text{NH}_3/\text{syngas}/\text{air}$ mixtures and $X_{\text{NH}_3} = 0.03$ – 1.0 in the $\text{NH}_3/\text{CO}/\text{air}$ mixtures measured using the high-pressure heat flux burner. The pressure exponents of S_L extracted from the experimental data are also obtained and compared to the simulated results. The NH_3 consumption and NO formation are numerically analyzed by four mechanisms. The main conclusions of this study are as follows.

- (1) It is found that the decrease of S_L as pressure increases or as X_{NH_3} increases is more evident in H_2 blended flames than in CO blended flames. The present measurements and predictions are in a good agreement for highly ammonia added conditions ($X_{\text{NH}_3} = 0.6$ – 1.0) and minorly ammonia added conditions ($X_{\text{NH}_3} = 0.0$ – 0.2), while the disparities are large for moderately ammonia added mixtures ($X_{\text{NH}_3} = 0.2$ – 0.6) which is due to the net ammonia burning rate ($f_{\text{NH}_3} \propto S_L \cdot X_{\text{NH}_3}$) is higher and introduces larger prediction uncertainty.

- (2) For $\text{NH}_3/\text{syngas}/\text{air}$ and $\text{NH}_3/\text{CO}/\text{air}$, Shrestha-Mech and UCSD-Mech lose its prediction accuracy at the lean and rich side while Han-Mech gives better predictions. The disparities increase from CO blending to H_2 blending conditions. While for $\text{NH}_3/\text{H}_2/\text{air}$, Stagni-Mech shows the better prediction for the lean side and Han-Mech, Shrestha-Mech agrees better with the experiments at the rich side but there is still a need for improvement.
- (3) The values of the pressure exponents β are similar for $\text{NH}_3/\text{syngas}/\text{air}$, $\text{NH}_3/\text{CO}/\text{air}$, and $\text{NH}_3/\text{H}_2/\text{air}$ flames experimentally and numerically. There are minima β at the intermediate ammonia content ($X_{\text{NH}_3} = 0.4\text{--}0.6$) which S_L is more sensitive to the pressure variation. The pressure exponent β may serve as an independent target for mechanism validation and improvement because the mechanism performance for S_L and β differs.
- (4) The difference of S_L due to the blending of NH_3 with syngas, CO, or H_2 is dominated by the chemical and diffusion effects. The adiabatic flame temperature is a significant factor for the similar pressure exponents β of the $\text{NH}_3/\text{syngas}/\text{air}$ flames with different H_2 to CO ratios, and for the different values of β for the fuel mixtures with different ammonia content, especially at the rich side.
- (5) Sensitivity and reaction pathway analyses show that ammonia chemistry plays a more important role at rich comparing to lean conditions, in affecting S_L especially via recombination reactions leading to N_2H_i which still possess large uncertainties. The higher ammonia decomposition rate and more emphasis of the N_2H_i reaction pathways in Han-Mech lead to the higher flame speed and reduction rate of S_L at rich conditions, agreeing better with the experimental $\text{NH}_3/\text{syngas}/\text{air}$ results. However, for $\text{NH}_3/\text{H}_2/\text{air}$ flame, the higher H_2NO , N_2O concentration and the increased role of hydrogen chemistry (H , HO_2 radicals) lead to higher flame velocity predicted by Stagni-Mech than by Han-Mech, which is also more consistent with experimental results at the lean and stoichiometric side.
- (6) Numerical results show that with the addition of hydrogen or carbon monoxide, the ammonia decomposition rate is significantly increased and the generation of NO is greatly strengthened. Shrestha-Mech predicts higher NO formation which is due to more NH_2 reacted with O to form HNO. The peak and tail concentration of NO_x (NO , N_2O and NO_2) are similar with the addition of hydrogen or carbon monoxide which is due to their similar flame temperature and thus similar ammonia decomposition rate; moreover, they play the same role in increasing the importance of H and O although their flame speeds differ a lot.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2020.08.004](https://doi.org/10.1016/j.combustflame.2020.08.004).

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